



Faculty of Technology & Engineering
Bachelor of Technology Programme
Computer Engineering
(B.Tech. CE)

ACADEMIC
REGULATIONS
&
SYLLABUS
(Choice Based Credit System)

Academic Year: 2023-2024

Devang Patel Institute of Advance Technology and Research

Vision

To be known as leading institute in the education and research by imparting knowledge which meets the requirements of digital age.

Mission

To engender globally competent engineers for the betterment of society.

Department of Computer Engineering

Vision

To emerge as an eminent research driven entity in Education in the field of computer engineering to meet the needs of the fast paced modern society.

Mission

To develop competent and responsible Computer Professionals for amelioration of the society at large.

THE PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1	To prepare the student(s) for successful career as an engineer, a corporate or a government professional, a scientist, an academician, a technocrat, an administrator and an entrepreneur.
PEO2	To make students demonstrate their abilities to adapt to a rapidly changing environment by having learning approach and apply new skills and new technologies to solve the problems.
PEO3	To create an ambience where the students are cared for in every aspect and motivated to become excellent working professionals who will continue to cherish their association with the organization as a whole, staff and colleagues.
PEO4	To provide continued professional development and lifelong learning throughout their career to inculcate strong teamwork and by installing lacking skills for the benefit of the society.
PEO5	To prepare the students to apply their technical skill(s) to analyse and design appropriate solution(s) with social consciousness and ethical values.

PROGRAM ARTICULATION MATRIX

(-Not Matching,1-Low,2,3-High)

Mission Statement	PEO1	PEO2	PEO3	PEO4	PEO5
To provide world-class educational activities	3	3	3	3	3
To deploy latest technologies	-	2	3	3	-
To foster research activities in consultation with academia and industry	3	2	2	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO1	Engineering knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO1	Apply good analytical, design and implementation skills required to formulate and solving computational problems.
PSO2	Excellent adaptability to function in multi-disciplinary work environment, good interpersonal skills in appreciation of professional ethics and societal responsibilities.

FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Bachelor of Technology Programmes
Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. *System of Education*

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. *Duration of Programme*

(i)	Undergraduate programme	(B. Tech.)
	Minimum	8 semesters (4 academic years)
	Maximum	16 semesters (8 academic years)

3. *Eligibility for admissions*

As enacted by Govt. of Gujarat from time to time.

4. *Mode of admissions*

As enacted by Govt. of Gujarat from time to time.

5. *Programme structure and Credits*

As per annexure – 1 attached

6. Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be Minimum 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

7.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

7.2 University Examination

7.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper and 100% for practical test or oral test or presentation by the student or a combination of any two or more of these.

7.2.2 **In order to earn the credit in a course a student has to obtain grade other than FF.**

7.3 Performance at Internal & University Examination

7.3.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	45%

7.3.2 A student failing to score 45% of the final examination will get a FF grade.

7.3.3 If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

- 8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	≥73 <80	≥66 <73	≥60 <66	≥55 <60	≥50 <55	≥45 <50	<45
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

- 8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.
- (iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.
- (iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- (v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

9. Awards of Degree

- 9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:
- 9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and
- 9.1.2 He should have cleared all internal and external evaluation components in every course; and
- 9.1.3 He should have secured a minimum CGPA of 4.5 at the end of the programme;
- 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.

9.2 The student who fails to satisfy minimum requirement of CGPA at the end of program will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	$\text{CGPA} \geq 7.5$
First class:	$\text{CGPA} \geq 6.0$
Second Class:	$\text{CGPA} \geq 5.0$

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/specialisation.

1.1.2 Programme Core (PC)

A ‘Core Course’ is a course which acts as a fundamental or conceptual base for Chosen Specialisation of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An ‘Elective Course’ is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

1.2.2 Programme Elective Course (PE):

A ‘Programme Elective Course’ is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time

1.2.3 Cluster Elective Course (CE):

An ‘Elective Course’ is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg.

Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.3 Non-Credit Course (NC) - AUDIT Course

A 'Non-Credit Course' is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student's Grade Sheet. Attendance and Course Assessment is compulsory for Non-Credit Courses

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CE ENGINEERING

CHOICE BASED CREDIT SYSTEM

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem 3	MA253	Discrete Mathematics and Algebra	4	0	4	4	30	70	0	0	100
	CE251	Java Programming	3	4	7	5	30	70	50	50	200
	CE260	Microprocessor and Computer Organization	3	2	5	4	30	70	25	25	150
	CE261	Data Structures & Algorithms	3	4	7	5	30	70	50	50	200
	XXXXX	University Elective - I	2			2	0	0	30	70	100
	CE244	Software Group Project-I	0	4	4	2	0	0	50	50	100
	HS121.02 A	Creativity Problem Solving & Innovation	0	2	2	2	0	0	30	70	100
			13	16	31	24	120	280	235	315	950
Sem 4	CE262	Data Communication & Networking	3	2	5	4	30	70	25	25	150
	CE263	Database Management System	3	4	7	5	30	70	50	50	200
	CE264	Design & Analysis of Algorithms	4	2	6	5	30	70	25	25	150
	CE265	Operating System	3	2	5	4	30	70	25	25	150
	CE259	Programming in Python	0	2	2	1	0	0	25	25	50
	XXXXX	University Elective - II	2			2	0	0	30	70	100
	CE255	Software Group Project-II	0	4	4	2	0	0	50	50	100
	HS111.02 A	Human Value & Professional Ethics	0	2	2	2	0	0	30	70	100
			15	18	33	25	120	280	260	340	1000

Note:

- **University Elective (UE):-** University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.
- **Cluster Elective (CT):-** Institutional Electives means common electives among a cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- **Programme Elective (PE):-**
- **Institute Elective (IE):-**

- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/program(s)

University Elective - I (UE - I)	University Elective - II (UE - II)
EC281.01: Introduction to MATLAB Programming	EC282.01: Prototyping Electronics with Arduino
CE281.01: Art of Programming	CE282.01: Web Designing
CL281.01: Environmental Sustainability and Climate Change	CL282.01: Basics of Environmental Impact Assessment
CL283: SDG Handprint Laboratory	EE287: MATLAB Programming
EE284: Python Programming	EE288: Maintenance of Household Apparatus
IT283: Web Designing & UI/UX	IT284: Data Visualization
ME281.01: Engineering Drawing	ME282.01: Material Science
PH233.01: Fundamentals of Packaging	PH238.02: Cosmetics in daily life
PD260.01: Basic Laboratory Techniques	NR261.01: Life Style Diseases & Management
NR251.01: First Aid & Life Support	PT192.01: Occupational Health & Ergonomics
PT191.01: Health Promotion and Fitness	CA225: Programming the Internet
CA224: Introduction to Web Designing	BM241: Health Care Management
BM231: Banking and Insurance	EC282.01: Prototyping Electronics with Arduino

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CE ENGINEERING

CHOICE BASED CREDIT SYSTEM

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem 5	CE354	Operating System	4	2	6	5	30	70	25	25	150
	CE355	Design & Analysis of Algorithms	4	2	6	5	30	70	25	25	150
	CE343	Software Engineering	3	2	5	4	30	70	25	25	150
	CE360	Machine Learning	3	2	5	4	30	70	25	25	150
	CEXXX	Programme Elective-I	2	4	6	4	30	70	50	50	200
	CE356	Software Group Project-III	0	4	4	2	0	0	50	50	100
	CE346	Summer Internship-I	0	0	0	3	0	0	75	75	150
	HSI31.02 A	Communication and Soft skills	0	2	2	2	0	0	30	70	100
			16	16	34	29	150	350	305	345	1150
Sem 6	CE357	Artificial Intelligence	3	2	5	4	30	70	25	25	150
	CE348	Information Security	4	2	6	5	30	70	25	25	150
	CE349	Theory of Computation	3	0	3	3	30	70	0	0	100
	CE358	Computer Networks	3	2	5	4	30	70	25	25	150
	CEXXX	Programme Elective-II	3	2	5	4	30	70	25	25	150
	CE359	Software Group Project-IV	0	4	4	2	0	0	50	50	100
	HSI32.02 A	Contributory Personality Development	0	2	2	2	0	0	30	70	100
			16	14	30	24	150	350	180	220	900

Note:

- **University Elective (UE):-** University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.
- **Cluster Elective (CT):-** Institutional Electives means common electives among a cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- **Programme Elective (PE):-**
- **Institute Elective (IE):-**

- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/program(s)

List of Programme Electives			
Code	Programme Elective - I (PE - I)	Code	Programme Elective - II (PE - II)
CE380	Advanced Java programming	CE374	Service Oriented Computing
CE381	Mobile Application Developments	CE378	Distributed System
CE382	Advanced web technology	CE379	Image Processing & Computer Vision
		CE383	Blockchain Technology

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CE ENGINEERING

CHOICE BASED CREDIT SYSTEM

	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem 7	CE442	Design of Language Processors	4	2	6	5	30	70	25	25	150
	CE443	Cloud Computing	3	2	5	4	30	70	25	25	150
	CE444	Internet of Things	3	2	5	4	30	70	25	25	150
	CEXXX	Programme Elective-III	3	2	5	4	30	70	25	25	150
	CE449	Big Data Analytics	3	2	5	4	30	70	25	25	150
	CE450	Software Group Project-V	0	4	4	2	0	0	50	50	100
	CE446	Summer Internship-II	0	0	0	3	0	0	75	75	150
			16	14	30	26	120	280	225	225	850
Sem 8	CE451	Software Project Major	0	36	36	18	0	0	250	350	600

Note:

- **University Elective (UE):-** University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.
- **Cluster Elective (CT):-** Institutional Electives means common electives among a cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- **Programme Elective (PE):-**
- **Institute Elective (IE):-**
- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/program(s)

List of Programme Electives		
ELECTIVES	Code	Programme Elective - III
	CE474	Blockchain Technology
	CE475	Wireless Communication & Mobile Computing
	CE476	Machine Learning

B. Tech. (Computer Engineering) Programme

SYLLABI (Semester – 3)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

MA253: DISCRETE MATHEMATICS AND ALGEBRA

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	0	-	4	4
Marks	100	0	-	100	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Predicate Calculus	08
2.	Relations and Lattice	10
3.	Graph Theory	12
4.	Recurrence Relations	05
5.	Abstract Algebra	09
6.	Linear Algebra	16
Total hours (Theory):		60

Detailed Syllabus:

1.	Predicate Calculus:	08 Hours	13%
	Revision: Propositions, connectives, converse, inverse, contrapositive, tautology, contradiction. Logical equivalence. Minimal functionally complete set of connectives. Principle conjunctive normal forms and Principle disjunctive normal forms. Predicate calculus using rules of inferences.		
2.	Relations and Lattice:	10 Hours	17%
	Revision of properties of relations on sets. Representations of relations: graphical and matrix representation. Equivalence relation, covering of a set, partition of a set. Partially ordered sets, totally ordered sets, Hasse diagram. Lattices, sub lattices. Properties of lattices (without proof). Complete lattices, bounded lattices, distributive lattices, complemented lattices and complemented distributive lattices.		
3.	Graph Theory:	12 Hours	20%

	Basic terminologies, Simple graph, Types of graphs. Degree of a vertex, matrix representations of graph. Path and connectivity. Eulerian and Hamiltonian graph. Subgraphs, spanning subgraphs, isomorphic graphs. Planar graphs. Matching in graphs. Graph coloring.		
4.	Recurrence Relations:	05 Hours	08%
	Solutions of recurrence relation by direct methods. Generating functions and solutions of recurrence relation.		
5.	Abstract Algebra:	09 Hours	15%
	Groupoid, semi group, monoid, group. Order of group, order of an element, Lagrange's theorem. Subgroup, cyclic subgroup, permutation group.		
6.	Linear Algebra:	16 Hours	27%
	Vector space: definition and examples. Subspaces. Linear combinations, linearly dependence and linearly independence. Basis and dimension of a vector space. Linear transformations. Null space and range of a linear transformation. Rank - nullity theorem. Isomorphisms.		

Student Learning Outcomes:

- At the end of the course the students would be able to

CO1	Develop logical argument using truth table and rules of inferences in predicate calculus.
CO2	Relation and types of relations define on sets and utilize it to construct Hasse diagram and lattices on sets.
CO3	Graph and types of the graphs and identify the real-world phenomena in terms of graph theory.
CO4	The concept of recurrence, generating functions and their applications in solving recurrence relations.
CO5	Different algebraic structures like groupoid, semi group, monoid, group, cyclic group and permutation group
CO6	Definition of vector space, concepts of the terms: linear span, linear independence, basis, dimension. Definition and properties of linear transformations, range and kernel of a linear transformation.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	-	1	-	-	-	-	-	-	-	2	-
CO2	3	2	-	-	-	-	-	-	-	-	-	-	3	-

CO3	3	2	1	-	2	-	-	-	-	-	-	-	3	1
CO4	3	-	-	-	1	-	-	-	-	-	-	-	3	-
CO5	3	1	-	-	-	-	-	-	-	-	-	-	2	-
CO6	2	1	-	1	-	-	-	-	-	-	-	-	2	-

Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Recommended Study Material:

❖ Text Books:

1. H. Anton and C. Rorres; Elementary Linear Algebra, Application version, Wiley Edition 2010.
2. Kenneth H. Rosen and Kamala Krithivasan. Discrete mathematics and its applications. Vol. 6. New York: McGraw-Hill, 1995.
3. Tremblay, Jean-Paul, and Rampurkar Manohar. Discrete mathematical structures with applications to computer science. New York: McGraw-Hill, 1975

❖ Reference Books:

1. Thomas H. Cormen, E. E. Leiserson, R. L. Rivest and C. Stein, Introduction to algorithms (Vol. 6). Cambridge: MIT press, 2001
2. Narsingh Deo, Graph theory with applications to engineering and computer science. Courier Dover Publications, 2016.
3. B. Kolman and R. C. Busby, Discrete Mathematical Structures for Computer Science, 2nd edition, Prentice-Hall, Englewood Cliffs, New Jersey 1987.
4. Swapan Kumar Sarkar, A Text Book of Discrete Mathematics, S. Chand and Co. New Delhi 2008.
5. D. S. Malik and Mridul K. Sen. Discrete mathematical structures: theory and applications. Course Technology, 2004.
6. D. F. McAllister and D. F. Stanat. Discrete Mathematics in Computer Science. Prentice-Hall, Inc. 1977.

❖ Web Links:

Lecture Notes:

1. <http://www.cs.yale.edu/homes/aspnes/classes/202/notes.pdf>
2. <http://home.iitk.ac.in/~aralal/book/mth202.pdf>
3. <https://web.stanford.edu/class/cs103x/cs103x-notes.pdf>
4. <https://www.cs.cornell.edu/~rafael/discmath.pdf>
5. <http://www-sop.inria.fr/members/Frederic.Havet/Cours/matching.pdf>
6. <http://www-sop.inria.fr/members/Frederic.Havet/Cours/coloration.pdf>

Video Lectures:

7. <http://www.nptelvideos.in/2012/11/discrete-mathematical-structures.html>

8. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-042j-mathematics-for-computer-science-fall-2010/video-lectures/>

CE251: JAVA PROGRAMMING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	4	-	7	5
Marks	100	100	-	200	

Pre-requisite courses:

- Basic programming skill.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Fundamental of Programming in Java	02
2.	Class Fundamentals	04
3.	Array & String Handling	02
4.	Inheritance, Interfaces & Packages	06
5.	Exceptions Handling	05
6.	Multithreaded Programming	07
7.	File I/O and NIO	07
8.	Java Collection Frameworks and Generics	12
	Total hours (Theory) :	45
	Total hours (Lab) :	60
	Total hours :	105

Detailed Syllabus:

1.	Fundamental of Programming in Java	02 Hours	04%
	History of Java, Basic overview of java, Bytecode, JVM Buzz-words, Application and applets, Constants, Variables & Data Types, Comment, Operators, Control Flow		
2.	Class Fundamentals	04 Hours	09%
	General form of class, Creating class Overloading methods, Constructor, Declaring Object, Returning objects, using objects as parameters, Assigning object reference variables,		

	Introducing Access control , Understanding static Introducing final, The finalize () method,The this keyword ,Garbage collection		
3.	Array & String Handling	02 Hours	04%
	Array basics, String Array, String class, StringBuffer and StringBuilder class, String Tokenizer Class and Object Class		
4.	Inheritance, Interfaces & Packages	06 Hours	14%
	Inheritance: Using super creating multilevel Hierarchy, method overriding, Dynamic method dispatch, abstract classes, Using final with Inheritance, Using Package: Defining package, Finding package and CLASSPATH, Access protection, Importing package, Interface: Defining Interface, Default Methods, Implementing Interface, Variables in Interface		
5.	Exceptions Handling	05 Hours	11%
	Exception types, Try ...Catch...Finally, Throw, Throws, creating your own exception subclasses		
6.	Multithreaded Programming	07 Hours	16%
	Life cycle of thread, thread methods, thread priority, thread exceptions, Implementing Runnable interface, Synchronization and Concurrency		
7.	File NIO	07 Hours	16%
	File and Directories, Byte streams and character streams, Random Access Files,NIO: Meta Data File Attributes Buffers, Channels, Recursive Operation.		
8.	Collection Framework and Generics	12 Hours	26%
	Collections of objects, Collections: Sets, Sequence, Map, Understanding Hashing, Use of Array List & Vector, Generics Class, Lamda Expression, Functional Reference, Method Reference, Optional Classes, Processing data with streams		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand and implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity and garbage collection for saving resources.
CO2	Demonstrate basic problem-solving skills: analyzing problems, modelling a problem as a system of objects, creating algorithms, and implementing models and algorithms in an object-oriented computer language (classes, objects, methods with parameters, abstract classes, interfaces, inheritance and polymorphism).
CO3	Use and develop Concurrency theory: progress guarantees, deadlock, live lock, starvation, linearizability.
CO4	Build and test program using new IO api and exception handling
CO5	Analyze and apply collection framework and generics to solve different data structure algorithms.
CO6	Understand and apply java new features

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	3	3	2	-	-	-	-	1	-	-	-	2	-
CO3	2	3	3	3	-	-	-	-	-	-	-	-	2	-
CO4	1	2	3	2	-	-	-	1	1	-	-	-	2	-
CO5	1	2	2	2	-	-	-	-	-	-	-	-	2	-
CO6	1	2	-	-	-	-	-	-	-	-	-	1	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text Books:**

1. Java: The Complete Reference, Ninth Edition by Herbert Schildt , Oracle Press.
2. Java 8 in Action: Lambdas, Streams, and Functional-style Programming by Alan Mycroft and Mario Fusco, Manning Publication

❖ **Reference Books:**

1. Thinking in Java, Bruce Eckel, Prentice Hall
2. Java: A Beginner's Guide (Sixth Edition) by Herbert Schildt , Oracle Press.
3. Core Java Volume I--Fundamentals (9th Edition) (Core Series) by Cay S. Horstmann, Prentice Hal

❖ **Web Materials:**

1. <https://docs.oracle.com>

CE260: MICROPROCESSOR AND COMPUTER ORGANIZATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Digital Electronics

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Register Transfer and Microoperations	07
2.	Basic Computer Organization and Design	09
3.	Central Processing Unit	04
4.	Pipeline and Vector Processing	02
5.	Computer Arithmetic	06
6.	Memory Organization	05
7.	8086,80186, 80286 Processor, 80386 Processors	10
8.	Current Era of Microprocessors	02
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Register Transfer and Microoperations	07 Hours	15%
	Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Microoperation, Logic Microoperations, Shift Microoperation, Arithmetic Logic Shift Unit.		
2.	Basic Computer Organization and Design	09 Hours	17%
	Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory Reference, Instructions, Input-Output and Interrupt, Complete Computer Description, Design of Basic Computer, Design of Accumulator Logic.		

3.	Central Processing Unit	04 Hours	10%
	Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes.		
4.	Pipeline and Vector Processing	02 Hours	06%
	Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processors.		
5.	Computer Arithmetic	06 Hours	15%
	Introduction: Binary, Octal, Decimal, Hexadecimal representation, Integer Numbers: Sign-Magnitude, 1's complement, 2's complement, Addition and Subtraction, Multiplication Algorithm.		
6.	Memory Organization	05 Hours	13%
	Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory.		
7.	8086, 80186, 80286 Processor, 80386 Processors	10 Hours	21%
	Architectural differences of 8086, 80186 and 80286 Processors, System Architecture, Registers, Memory management: Segment Translation, Page Translation, Combining Segment and Page Translation.		
8.	Current Era of Microprocessors	02 Hours	03%
	Comparison of AMD and Intel Architecture, Features of current era of AMD and Intel processors; Tick-Tock: manufacturing pattern of Intel.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand and Design of Arithmetic circuit, Logic circuit, Shift circuit and Control Unit circuit by using elements of digital logic.
CO2	Visualize and understand the working of CPU, different instruction formats, addressing modes, pipeline and vector processing. Evaluate the performance of pipeline approach.
CO3	Understand and apply r's and r-1's complement in Addition, Subtraction and Multiplication of signed and unsigned numbers.
CO4	Describe the memory hierarchy and requirements of different memory. Compare and evaluate different cache memory mapping techniques.
CO5	Understand and differentiate processors development from history to current era.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
--	-----	-----	-----	-----	-----	-----	-----	-----	-----	------	------	------	------	------

CO1	3	3	3	-	-	-	-	1	1	-	-	1	2	-
CO2	3	3	3	1	-	-	-	1	1	-	-	-	2	-
CO3	2	2	3	-	-	-	-	1	1	-	-	-	2	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-	2	-
CO5	3	3	3	3	-	-	-	-	-	-	-	-	3	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Computer System Architecture, Morris Mano (3rd Edition) Prentice Hall.
2. 80386 Programmer's reference manual from MIT.
3. Microprocessors and Interfacing: Experiments Manual: Programming and Hardware by Douglas V. Hall.

❖ Reference book:

1. William Stalling, Computer Organization & Architecture-Designing for Performance, Pearson Prentice Hall (8th Edition).
2. A.S. Tananbum, Structured Computer Organization, Pearson Publisher.
3. The Essentials of Computer Organization and Architecture Linda Null, Julia Lobur.
4. John P Hayes, Computer Architecture & Organization, McGraw-Hill.
5. Computer Architecture: Pipelined and Parallel Processor Design Michael J. Flynn (4th edition).

❖ Web Material:

1. Computer Organization and Architecture: A Pedagogical Aspect: https://onlinecourses.nptel.ac.in/noc21_cs37/course
2. <https://css.csail.mit.edu/6.858/2014/readings/i386.pdf> (80386 Programmer Reference Material)

❖ Software:

1. 8085 & 8086 Simulator.
2. Logisim : <http://www.cburch.com/logisim/>

3. Little Man Computer

- a. <https://peterhigginson.co.uk/LMC/>
- b. <https://www.101computing.net/LMC/>
- c. <https://www.youtube.com/watch?v=fXMCnzdNemc>
- d. <https://www.youtube.com/watch?v=sEFnRDgkaWA>

CE261: DATA STRUCTURE & ALGORITHMS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	4	-	7	5
Marks	100	100	-	200	

Pre-requisite courses:

- Programming Language

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Data Structure	04
2.	Linear Data Structure	12
3.	Non-Linear Data Structure	16
4.	Sorting	10
5.	Searching	01
6.	Dictionaries	02
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Introduction	04 Hours	08%
	Introduction to data structure (Types of data structure), Introduction to algorithms. Algorithm Analysis and Big O notation, Memory representation of Array: Row Order and Column Order, Abstract Data Types (ADT)		
2.	Linear Data Structure	12 Hours	27%
	Stack: Operations: push, pop, peep, change, Applications of Stack: Recursion: Recursive Function Tracing, Principles of recursion, Tail recursion, Removal of Recursion, Tower of Hanoi, Conversion: Infix to Postfix, Infix to Prefix.		

	Evaluation: Prefix and Postfix expression, Queue Simple Queue: Insert and Delete operation, Circular Queue: Insert and Delete operation, Concepts of: Priority Queue, Double-ended Queue, Applications of Queue, Linked List: Memory Representation of LL, Singly Linked List, Doubly Linked List, Circular Linked List, Applications of Linked List		
3.	Non-Linear Data Structure	16 Hours	36%
	Tree: Tree Concepts, Tree Traversal Techniques: Pre-order, Post-order and In-order (Recursive and Iterative), Binary Search Tree: Iterative and Recursive, Balanced Trees (AVL Trees, Applications of Tree, Skip list Heaps: Priority queues and Binary Heaps Graph: Graph concepts, Memory Representation of Graph, BFS and DFS, Applications of Graph		
4.	Sorting	10 Hours	23%
	Sorting (concepts, Selection Sort, Bubble Sort, Merge Sort, Radix Sort, Insertion Sort, Heap Sort, Quick Sort, Counting sort, Topological sort)		
5.	Searching	01 Hours	02%
	Sequential Search, Binary Search		
6.	Dictionaries	02 Hours	04%
	Hashing, Hashing Functions, Collision-Resolution Techniques, Applications		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand and Implement Algorithms and core Data Structures such as stack, queue, hash table, priority queue, binary search tree and graph in programming language.
CO2	Analyse data structures in storage, retrieval and computation of ordered or unordered data.
CO3	Compare alternative implementations of data structures with respect to demand

	and performance.
CO4	Describe and evaluate the properties, operations, applications, strengths and weaknesses of different data structures.
CO5	Apply and select the most suitable data structures to solve programming challenges.
CO6	Discover advantages and disadvantages of specific algorithms.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	2	-	-	-	-	-	-	-	-	-	3	-
CO2	-	2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	-	3	3	3	-	-	-	-	-	-	-	-	2	-
CO4	-	1	-	1	2	-	-	-	-	-	-	-	2	-
CO5	2	2	2	2	-	-	-	-	-	-	-	2	3	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	2	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. An Introduction to Data Structures with Applications, Jean-Paul Tremblay, Paul G. Sorenson, McGraw-Hill.
2. Data structure with C, Lipschutz, TMH
3. Introduction to Algorithms: Cormen, Leiserson, Rivest and Stein: Prentice Hall of India
4. Data Structures and Algorithms: Aho, Hopcroft and Ullmann: Addison Wesley.

❖ Reference book:

1. Classic Data structures, D.Samanta, Prentice-Hall International.
2. Data Structures using C & C++, Ten Baum, Prentice-Hall International.

3. Data Structures: A Pseudo-code approach with C, Gilberg & Forouzan, Thomson Learning.
4. Fundamentals of Data Structures in C++, Ellis Horowitz, Sartaj Sahni, Dinesh Mehta, W. H. Freeman.
5. “A Practical Introduction to Data Structures and Algorithm Analysis” by Clifford A. Shaffer
6. Data Structures and Algorithm in Java: Goodrich and Tamassia: John Wiley and Sons.

❖ **Web material:**

1. <http://www.leda-tutorial.org/en/official/ch02s02s03.html>
2. <http://www.leda-tutorial.org/en/official/ch02s02s03.html>
3. <http://www.softpanorama.org/Algorithms/sorting.shtml>

❖ **Software:**

1. Wireshark
2. Cisco Packet Tracer

CE244: SOFTWARE GROUP PROJECT-I

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	0	4	-	4	2
Marks	0	100	-	100	

Pre-requisite courses:

- Programming Language, Software Engineering.

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - a. Final Project Report
 - b. Project Setup file with Source code
 - c. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 60

Total hours : 60

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Create enhanced employment by moulding the students with higher technical skill.
CO2	Promote creative thinking, to provide hands on actual technology.
CO3	Handle software project and get use to software development processes.
CO4	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Improve communication and presentation skill.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	-	-	-	-	-	-	-	1	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	2	-	-	-	2	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO5	-	-	1	-	3	2	-	-	-	3	-	-	-	2
CO6	-	-	-	-	-	-	3	1	-	3	-	2	-	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:❖ **Reference book:**

1. Books, Magazines ,Journals & online course platforms of related topics

❖ **Web material:**

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <https://www.udemy.com/>

5. <https://www.udacity.com/>
6. <https://nptel.ac.in/course.html>
7. <https://www.futurelearn.com/>

❖ **Software:**

1. PYTHON/MATLAB
2. PHP
3. ANDROID/IOS

HS121.02 A: CREATIVITY, PROBLEM SOLVING AND INNOVATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

Creative Problem Solving

<https://www.coursera.org/learn/creative-problem-solving>

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	06
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1.	Introduction to Creativity, Problem Solving and Innovation	6 Hours	20%
	Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills		
2.	Questioning, Learning and Visualization	6 Hours	20%
	Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking		
3.	Creative Thinking and Problem Solving	6 Hours	20%

	Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving		
4.	Logic, Language and Reasoning	6 Hours	20%
	Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning		
5.	Contemporary Issues and Practices in Creativity and Problem Solving	6 Hours	20%
	Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	-	1	1	-	-	-	2	-	1	1	-
CO2	-	3	2	1	-	-	-	1	1	1	-	1	-	1
CO3	-	2	2	1	-	-	-	-	-	-	-	-	-	-
CO4	1	-	1	-	-	-	2	-	-	-	-	-	1	-
CO5	-	1	-	1	-	-	-	-	1	3	-	1	-	1
CO6	-	-	-	2	-	-	-	-	-	-	2	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossey-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

❖ Reference book:

1. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
2. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
3. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
4. Tony Buzan, How to Mind Map, Thorsons Publication 2002
5. Scott Berkun, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
6. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
7. Ira Flatow, The all Laughed, Harper Publication 1992
8. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

❖ Additional Readings:

1. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
4. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
5. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
9. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, Scott G Isaksen, K Brainstead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
11. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
13. Howard Gardner, Creating minds, Basic Books Publication 1993

14. MihalyCsikzentmihalyi, Creativity–Flow and Psychology of Discovery and Invention,Harper Publication 1996
15. Martin Gerdner,W. H.,Ahal Insight,Freeman Publication 1978
16. Paul Sloane,Test Your Lateral Thinking IQ,Sterling Publication1994
17. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles,Sterling Publication 1996

❖ **Web material:**

1. Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

B. Tech. (Computer Engineering) Programme

SYLLABI (Semester – 4)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE262: DATA COMMUNICATION & NETWORKING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Digital Electronics

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction and Basic Concepts	02
2.	Network Model	02
3.	Signals	03
4.	Digital Transmission	04
5.	Multiplexing	03
6.	Data Transmission	04
7.	Switching	03
8.	Media Access Control	05
9.	Network Routing	08
10	Transport Layer	08
11.	Application Layer	03
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Introduction and Basic Concepts	02 Hours	4%
	Data Communication, Networks, Network Types		
2.	Network Model	02 Hours	4%

	Protocol Layering, TCP/IP Protocol Suites, The OSI Model		
3.	Signals	03 Hours	6%
	Data and Signals, Periodic Analog Signal, Digital Signal, Transmission Impairment, Data Rate limits, Performance		
4.	Digital Transmission	04 Hours	9%
	Digital to Digital Conversion, Analog to Digital Conversion, Transmission Modes,		
5.	Multiplexing	03 Hours	6%
	Frequency division Multiplexing, Wave length division Multiplexing, Time division Multiplexing		
6.	Data Transmission	04 Hours	9%
	Guided Media, Unguided Media		
7.	Switching	03 Hours	9%
	Circuit Switching, Packet Switching, Structure of Switch		
8.	Media Access Control	05 Hours	11%
	Random Access, Controlled Access, Channelization		
9.	Network Routing	08 Hours	18%
	Unicast Routing, What's Inside a Router, The Internet Protocol (IP): Forwarding and Addressing in the Internet, Routing Algorithm(SS)		
10.	Transport Layer	08 Hours	18%
	Introduction and Transport-Layer Services, Multiplexing and De-multiplexing, Connectionless Transport: UDP & Building Blocks of UDP, Principles of Reliable Data Transfer, Connection-Oriented Transport: TCP, Principles of Congestion Control & Building Blocks of TCP		
11.	Application Layer	03 Hours	6%
	Principles of Network Applications, The Web and HTTP, File Transfer: FTP, SMTP, DNS, Optimizing Application Delivery		

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand and identify different physical layer transmission fundamentals such as types of signals, transmission, multiplexing, types of medium and modulation.
CO2	Understand working of application layer protocols.
CO3	Understand and Analyse transport layer services and its impacts on data rate.
CO4	Understand and differentiate functionality of existing network routing protocols.
CO5	Understand existing different medium access protocols and evaluate for adoption for future networking.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	1	-	-	-	-	-	-	-	2	-	-
CO2	1	2	1	1	2	-	-	-	-	-	-	2	1	1
CO3	3	2	2	1	2	-	-	-	-	1	-	2	1	1
CO4	2	3	1	1	1	-	-	-	-	2	-	2	-	-
CO5	3	2	3	1	3	-	-	-	1	1	-	3	1	1
CO6	3	3	3	2	3	-	-	-	2	2	-	3	3	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Practical Content:

Ethernet Protocol, Standard Ethernet, Fast Ethernet, Gigabit Ethernet, Wireless characteristics and access control, IEEE 802.11 project.

Recommended Study Material:

❖ Text book:

1. Data communication & Networking, Behrouz Forouzan, McGraw-Hill
2. Computer Networking: A Top-Down Approach James F. Kurose, University of Massachusetts, Amherst Keith W. Ross, Polytechnic University, Brooklyn

❖ Reference book:

1. Data and Computer Communications, William Stallings, Prentice Hall
2. Computer Network, Andrew S. Tanenbaum, Fourth Edition, Prentice Hall

❖ **Web material:**

1. www.wikipedia.org
2. <http://www.webopedia.com>

❖ **Software:**

1. Wireshark
2. Cisco Packet Tracer
3. NS-3
4. Ubuntu Server Installation

CE263: DATABASE MANAGEMENT SYSTEM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	4	-	7	5
Marks	100	100	-	200	

Pre-requisite courses:

- Data Structure

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introductory concepts of DBMS	02
2.	Relational Model	04
3.	Entity-Relationship model	05
4.	Formal Relational Query Languages	04
5.	Relational Database Design	07
6.	Transaction & Recovery Management	06
7.	Advanced Transaction Processing	05
8.	Database Security	06
9.	Indexing and Hashing	06
	Total hours (Theory) :	45
	Total hours (Lab) :	60
	Total hours :	105

Detailed Syllabus:

1.	Introductory concepts of DBMS	02 Hours	06%
	Introduction and applications of DBMS, Purpose of database, Data Independence, Database System architecture-levels, Mappings, Database users and DBA		
2.	Relational Model	04 Hours	10%
	Structure of Relational Databases, Database Schema, Schema Diagram, Domains , Relations, Relational Query		

	Languages, Relational Operations		
3.	Entity-Relationship model	05 Hours	12%
	Basic concepts, Design process, Constraints, Keys, Design issues, E-R diagrams, Weak Entity Sets, Extended E-R features- Generalization, Specialization, Aggregation, Reduction to E-R database schema		
4.	Formal Relational Query Languages	04 Hours	10%
	The relational Algebra, The Tuple Relational Calculus, The Domain Relational Calculus		
5.	Relational Database design	07 Hours	15%
	Functional Dependency–definition, Trivial and Non-Trivial FD, Closure of FD set, Closure of attributes, Irreducible set of FD, Normalization – 1NF, 2NF,3NF, Decomposition using FD- Dependency Preservation		
6.	Transaction & Recovery Management	06 Hours	12%
	Transaction concepts, Properties of Transactions, Serializability of transactions, Testing for Serializability, System recovery, Two- Phase Commit protocol, Recovery and Atomicity, Log-based recovery, Concurrent executions of transactions and related problems, Locking mechanism, Solution to Concurrency Related Problems, Deadlock, Two-phase locking protocol, Intent locking		
7.	Advanced Transaction Processing	05 Hours	10%
	Transaction-Processing Monitors, Transactional Workflows, Main-Memory Databases, Real-Time Transaction Systems, Long-Duration Transactions		
8.	Database Security	06 Hours	10%
	Views - What are views for?, View retrievals, View updates, Snapshots (a digression), Materialized view, Security – Security and Authentication, authorization in SQL, Data encryption, Missing Information - An overview of the 3VL approach		
9.	Indexing and Hashing	06 Hours	12%

	Basic Concepts, Ordered Indices, B+-Tree Index Files, B+-Tree Extensions, Multiple-Key Access, Static Hashing, Dynamic Hashing, Comparison of Ordered Indexing and Hashing, Bitmap Indices, Index Definition in SQL		
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Course Outcomes (COs):

At the end of the course, students will be able to

CO1	Apply the concepts of engineering i.e. collecting data, organize the data in the systematic form, and arrange the data in a computational way and applying mathematics formation.
CO2	Analyse how data are stored and maintained using data models. Ready to assimilate the concept of data abstraction and design queries using SQL. Identify how data is represented in the relational model and create relations using SQL language
CO3	Identify and evaluate the constructs in the E-R model and issues involved in developing an E-R diagram. Convert an E-R diagram into a relational database schema. Declare and enforce integrity constraints on database using a state-of-art RDBMS.
CO4	Produce aggregate operators to write SQL queries which are not expressible in relational algebra. “More mathematical” notation may apply and also used in research and other venues. Combining these concepts allows production of sophisticated queries.
CO5	Decompose un-normalized tables into normalized compliant tables. Design and implement a normalize database schema for a given problem-domain. Produce strategies to minimise risks of security breaches in a range of network environments and data storage systems. Compute retrieval time and concluding with suitable indexing technique.
CO6	Compare transactions and their properties with (ACID) and without ACID. Apply locking protocol to ensure isolation. Develop logging technique to ensure atomicity and durability. Design a logical view which can be used for analytical tasks. Develop practical experience of the design and implement scalable, secure databases.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	1	2	1	1	-	-	-	1	2	-
CO2	3	3	3	2	3	1	2	-	-	-	-	3	2	1
CO3	3	3	3	3	2	2	1	3	2	1	-	2	3	2
CO4	3	3	1	3	1	1	-	2	2	-	-	2	2	2
CO5	3	3	3	2	3	1	-	2	2	1	-	3	2	2
CO6	3	3	2	2	-	2	1	3	1	-	-	3	3	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. Database System Concepts, Abraham Silberschatz, Henry F. Korth & S. Sudarshan, McGraw Hill.
2. An introduction to Database Systems, C J Date, Addison-Wesley

❖ Reference book:

1. “Fundamentals of Database Systems”, R. Elmasri and S.B. Navathe, the Benjamin / Cumming Pub. Co
2. SQL, PL/SQL the Programming Language of oracle, Ivan Bayross, BPB Publications
3. Oracle: The Complete Reference, George Koch, Kevin Loney, TMH /oracle press

❖ Web material:

1. <http://www.sql.org>
2. <http://www.w3schools.com>
3. <http://www.sqlcourse.com>

❖ **Software:**

1. Oracle 10g
2. SQL Lite
3. Live SQL
4. Firebase
5. Squirrel SQL
6. Postgre SQL

CE264: DESIGN & ANALYSIS OF ALGORITHMS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	0	6	5
Marks	100	50	0	150	

Pre-requisite courses:

- Data Structure and Algorithms
- Programming language
- Discrete Mathematics

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	14
2.	Greedy Algorithm	16
3.	Dynamic Programming	14
4.	Divide and Conquer Algorithm	06
5.	Searching Algorithms and String Matching Algorithms	06
6.	Complexity Theory	04
	Total hours (Theory):	60
	Total hours (Lab):	30
	Total hours:	90

Detailed Syllabus:

1.	Introduction	14 Hours	24%
	Introduction to Algorithms, Analysis and Design Techniques of Algorithm, Model for Analysis of Algorithm: Random Access Machine (RAM), Algorithm Analysis Techniques: Mathematics, Empirical and Asymptotic Analysis: Big-O, Big-Theta and Big-Omega, worst, average and best case analysis Complexity Analysis of Iterative Algorithm: Primitive Operations, Analysing control statements		

	Complexity Analysis techniques for Recurrence Relations: substitution method, Iterative Method, Recursion Tree Method, Master's Theorem Basic Sorting Algorithms with best case, average case and worst case analysis: Selection Sort, Insertion Sort, Bubble Sort		
2.	Greedy Design Technique	16 Hours	26%
	Greedy choice and optimal substructure property of Greedy approach, Problem Solving using greedy: making change, fractional knapsack, Job Scheduling, Huffman coding, Optimal Merge Pattern, Graph Coloring Algorithm. Minimum spanning trees -Prims and Kruskal algorithm. Graph algorithms: Breath First Search, Depth First Search, Shortest Path Algorithms: Dijkstra's algorithm, Floyd-Warshall Algorithm, Bellman-Ford Algorithm.		
3.	Dynamic Programming	14 Hours	24%
	Dynamic Programming: The Principle of Optimality, Dynamic Programming using memorization and bottom up approach, problem solving using Dynamic Programming: Calculating the Binomial Coefficient, Assembly Line-Scheduling, Knapsack Problem, Matrix Chain Multiplication, Sum of Subset Problem, Longest Common Subsequence		
4.	Divide and Conquer Approach	06 Hours	10%
	Binary Search, Sorting Algorithms with best, average and worst case analysis: Merge Sort and Quick Sort, Matrix Multiplication		
5.	Searching Algorithms and String Matching Algorithms	06 Hours	10%
	Backtracking –The Knapsack Problem; The Eight Queens problem, Sum of Subset Problem Branch and Bound –The Assignment Problem, The Knapsack Problem String Matching Algorithms: The naïve string-matching algorithm, The Rabin-Karp algorithm, KMP Algorithm for Pattern Searching, Boyer–Moore string-search algorithm		
6.	Complexity Theory	4 Hours	6%

	The class P and NP Problem, Polynomial reduction, NP-Completeness Problem, NP-Hard problems		
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Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Analyse how to analyse algorithms and estimate their best case, worst-case and average-case behaviour (in easy cases)
CO2	Describe the greedy paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm and analyse them.
CO3	Describe the dynamic-programming paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm and analyse them.
CO4	Describe the divide-and-conquer paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm. Derive and solve recurrences describing the performance of divide-and-conquer algorithms.
CO5	Understand and apply various graph algorithms for finding shortest path and minimum spanning tree.
CO6	Understand and apply string matching algorithms for finding the pattern from the text.
CO7	Understand a problem as computationally tractable or intractable, and discuss strategies to address intractability.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO0	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	-	-	-	-	-	-	-	-	1	-
CO2	2	2	-	-	-	-	-	-	-	-	-	2	2	-
CO3	3	3	3	3	2	-	-	-	-	-	-	2	2	-
CO4	2	3	3	1	-	-	-	-	-	-	-	-	2	-
CO5	1	-	1	-	-	-	-	-	-	-	-	2	1	1
CO6	3	1	-	-	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald Rivest and Clifford Stein, MIT Press

❖ **Reference book:**

1. Fundamental of Algorithms by Gills Brassard, Paul Bratley, Pentice Hall of India.
2. Fundamental of Computer Algorithms by Ellis Horowitz, Sartazsahni and sanguthevar Rajasekarm, Computer Sci.P.
3. Design & Analysis of Algorithms by P H Dave & H B Dave, Pearson Education.

❖ **Web material:**

1. <http://highered.mcgraw-hill.com/sites/0073523402/>

CE265: OPERATING SYSTEM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	4
Marks	100	50	0	150	

Pre-requisite courses:

- Introduction to computer and computer organization.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	02
2.	Process Management	06
3.	Process Synchronization	06
4.	Deadlocks	04
5.	Memory Management	10
6.	Secondary Storage and Input Output Management	04
7.	File Systems	04
8.	System Protection	03
9.	Unix/Linux File System	04
10.	Epilogue	02
	Total hours (Theory):	45
	Total hours (Lab):	30
	Total hours:	75

Detailed Syllabus:

1.	Introduction	02 Hours	05%
	What is an OS? OS structure, System Calls, Functions of OS, Evolution Of OS, OS design consideration for Multiprocessor and Multicore		
2.	Process Management	06 Hours	13%
	Process: What is Process? Process Control Block, Process States, Process API: fork(), exec() and wait() system calls, Process Scheduling, Threads: Types of Threads, Multithreading model, Scheduler		

	activation		
3.	Process Synchronization	06 Hours	13%
	Race Conditions, Critical Section, Peterson's Solution, hardware primitives for synchronization, Mutex Locks, Semaphores, Monitors and Classical problems in synchronization(Producer-Consumer, Reader's & Writer Problem, Dining Philosopher Problem).		
4.	Deadlocks	04 Hours	9%
	Modelling, Characterization, Prevention and Avoidance, Detection and Recovery		
5.	Memory Management	10 Hours	22%
	Address Space, Address Translation, Multiprogramming with Fixed and variable partitions, Swapping, Segmentation, Free Space Management, Paging: Principle Of Operation, Page Allocation, H/W Support For Paging. Multilevel paging, Virtual Memory: Concept, Performance of Demand Paging, Page Replacement Algorithms, Thrashing and Working Sets.		
6.	Secondary Storage and Input Output Management	04 Hours	09%
	I/O Devices: Device Controllers, Device Drivers, Direct Memory Access. Goals of The I/O S/W, Interrupt Handler, Device Independent I/O Software Disks: Disk architecture, Disks Scheduling Algorithm, RAID		
7.	File Systems	04 Hours	09%
	File: File Naming, File attributes, File Operations, File Structure, File Types, Access Methods, Directories: Structure, Pathnames, Directory Operations. File System Implementation: partitioning and mounting, Allocation methods: Contiguous Allocation, Linked Allocation, Indexed Allocation.		
8.	System Protection	03 Hours	07%
	Goals of protection, Principles of protection, Domain of protection, Access matrix, Implementation of Access matrix, Access control		
9.	Unix/Linux File System	04 Hours	09%
	Buffer Cache, Inodes, The system calls - ialloc, ifree, namei, alloc and free. Mounting and Unmounting , files systems, Network File systems. EXT file system in linux		
10.	Epilogue	02 hours	04%
	Pointers to advanced topics: distributed OS, multimedia OS, embedded OS, real-time OS, OS for multiprocessor machines,		

	research problems in OS		
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Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Analyze the basic structure of OS and architectural components involved in OS design
CO2	Evaluate the performance of various process scheduling algorithms after understanding various process states and its mechanism to manage
CO3	Evaluate the requirements of process synchronization and coordination in contemporary operating system.
CO4	Compare the various mechanisms to manage memory by applying memory management algorithms and schemes along with management of input/output devices
CO5	Compare the various mechanisms to manage files by applying file allocation methods along with security aspects
CO6	Analyze the administration of Linux operating system

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO0	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	2	-	-	-	-	-	-	1	-	-
CO2	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO3	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO4	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO5	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO6	2	2	-	-	2	-	-	-	-	-	-	1	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Operating System Concepts Avi Silberschatz, Peter Baer Galvin, Greg Gagne, Ninth Edition, Wiley
2. Operating Systems Internals and Design Principles, William Stallings, Seventh Edition, Prentice Hall

❖ **Reference book:**

1. Operating Systems: Three Easy Pieces by Remzi Arpaci-Dusseau and Andrea Arpaci-Dusseau (available online for free)
2. Modern Operating Systems -By Andrew S. Tanenbaum, Third Edition PHI
3. Operating Systems, D.M. Dhamdhare, TMH
4. Unix System Concepts & Applications, Sumitabha Das, TMH
5. Unix Shell Programming, Yashwant Kanitkar, BPB Publications

❖ **Web material:**

1. NPTEL course: <https://nptel.ac.in/courses/106/106/106106144/>
2. Simulator: <https://cpudemo.azurewebsites.net/#/>

❖ **Software:**

1. Simulator: OS Sim (OS Concepts Simulator)

CE259: PROGRAMMING IN PYTHON

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	1
Marks	-	50	-	50	

Pre-requisite courses:

- High level language (C/C++/Java)
- Web Programming

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basics of Python	02
2	Data Structures: Lists, Tuples, Dictionaries and Strings	04
3	Control structures and Function	04
4	Modules and Scoping Rules	02
5	Exceptions Handling	04
6	Magic Methods, Properties, and Iterators	04
7	Object Oriented Programming	06
8	Regular Expression and File Handling	04
	Total hours (Theory) :	0
	Total hours (Lab) :	30
	Total hours :	30

Detailed Syllabus:

1.	Basics of Python	02 Hours	07 %
	Using the Python Interpreter, Variables, Identifiers and Keywords, Numbers and Expressions		
2.	Data Structures: List, Tuples, Dictionaries and Strings	04 Hours	13 %
	Common Sequence Operations: Indexing, Slicing, Adding Sequences, Multiplication, Membership, Length, Minimum, and Maximum, Using Lists as Stacks, Using Lists as Queues, List Comprehensions, Nested List Comprehensions, the del statement,		

	Tuples and Sequences, Sets, Dictionaries, Comparing Sequences and Other Types, Basic String Operations		
3.	Control Structures and Functions	04 Hours	13%
	Conditional Branching: if Statements, break and continue Statements, and else Clauses on Loops, pass Statements Loops: while Loops, for Loops, Defining Functions, More on Defining Functions: Default Argument Values, Keyword Arguments, Arbitrary Argument Lists, Unpacking Argument Lists, Lambda Expressions, Documentation Strings, Function Annotations		
4.	Modules and Scoping Rules	02 Hours	07%
	Executing modules as scripts, The Module Search Path, “Compiled” Python files, Packages: Importing * From a Package, Intra-package References, Packages in Multiple Directories		
5.	Exception Handling	04 Hours	13%
	Syntax Errors, Exceptions, Handling Exceptions, Raising Exceptions, User-defined Exceptions, Defining Clean-up Actions, Predefined Clean-up Actions		
6.	Magic Methods, Properties and Iterators	04 Hours	13%
	Constructors, Item Access: The Basic Sequence and Mapping Protocol, Properties: The property Function, Static Methods and Class Methods, __getattr__, __setattr__, and Friends, Iterators, Generators, Generator Expressions		
7.	Object Oriented Programming	06 Hours	20%
	Python Scopes and Namespaces, Class Definition, Class Objects, Instance Objects, Method Objects, Class and Instance Variables, Inheritance, Multiple Inheritance, Private Variables, Polymorphism, Using Properties to Control Attribute Access, Creating Complete Fully Integrated Data Types		
8.	Regular Expression and File Handling	04 Hours	14%
	What is a regular expression?, Regular expressions with special characters, Regular expressions and raw strings, Extracting matched text from strings, Substituting text with regular expressions, Writing and Reading Binary Data, Writing and Parsing Text Files, Iterating		

	over File Contents, Writing and Parsing XML Files, Random Access Binary Files		
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Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Interpret the fundamental python syntax, semantics and fluent in the use of python control flow statements. Express proficiency in the handling of strings and functions.
CO2	Determine the methods to create and manipulate python programs by utilizing the data structures like lists, dictionaries, tuples and sets.
CO3	Identify the commonly used operations involving file systems and regular expressions.
CO4	Articulate the Object-Oriented Programming concepts such as encapsulation, inheritance and polymorphism as used in Python along with magic methods.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	3	-	-	-	-	-	-	1	-	1
CO2	3	3	2	1	3	-	-	-	-	-	-	-	2	-
CO3	2	2	1	1	3	-	-	-	-	-	-	-	2	-
CO4	3	2	2	2	3	-	-	-	-	-	-	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Magnus Lie Hetland, "Beginning Python From Novice to Professional", Third Edition, Apress, 2017
2. Nigel George, "Mastering Django: Core" Packt Publishing, 2016

❖ **Reference book:**

1. David Beazley, Brian K. Jones, "Python Cookbook", 3rd edition, O'REILLY, 2016
2. Brett Slatkin, "Effective Python: 59 Specific Ways to Write Better Python", Novatec, 2016
3. Allen Downey, "Think Python: How to Think Like a Computer Scientist", Green Tea Press, 2015
4. Mark Lutz "Learning Python", 4th Edition, O'REILLY, 2016
5. Arun Ravindran, Aidas Bendoraitis, Samuel Dauzon, "Django: Web Development with Python", Packt Publishing, 2016

❖ **Web material:**

1. <https://www.python.org/>
2. <http://www.diveintopython3.net/>
3. <https://developer.mozilla.org/en-US/docs/Learn/Server-side/Django>
4. <https://www.fullstackpython.com/django.html>

❖ **Software:**

1. Python IDLE
2. Anaconda Python
3. PyCharm

CE255: SOFTWARE GROUP PROJECT-II

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	4	-	4	2
Marks	-	100	-	100	

Pre-requisite courses:

- Programming Language, Software Engineering.

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - d. Final Project Report
 - e. Project Setup file with Source code
 - f. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 60

Total hours : 60

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify problems present in society by surveying variety of domains and convert in project definition.
CO2	Explore new ideas and techniques to solve it. Create, select and apply appropriate techniques, resources, modern engineering and IT tools to solve problem.
CO3	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO4	Apply ethical principles and commit to responsibilities and norms of the project.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Apply engineering and management principles to achieve project goal.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	2	-	-	-	-	-	-	-	-	-	-
CO2	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	-	-	-	-	-	-	3	2	-	-	-	-	-
CO5	-	-	1	-	3	2	-	-	-	3	-	-	-	-
CO6	3	2	2	1	2	1	-	-	2	3	-	1	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:❖ **Reference book:**

1. Books, Magazines ,Journals & online course platforms of related topics

❖ **Web material:**

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <https://www.udemy.com/>
5. <https://www.udacity.com/>

6. <https://nptel.ac.in/course.html>

7. <https://www.futurelearn.com/>

❖ **Software:**

1. ASP.NET

2. PYTHON/MATLAB

3. PHP

4. ANDROID/IOS

HS111.02 A: HUMAN VALUES & PROFESSIONAL ETHICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/01	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

- Ethical Leadership through Giving Voice to Values
<https://www.coursera.org/learn/uva-darden-giving-voice-to-values?skipBrowseRedirect=true>

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Elements and Principles of Values	08
3.	Applied Ethics	08
4.	Value, Ethics & Global Issues	05
5.	Contemporary Issues in Values and Ethics	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	Introduction to Values and Ethics	5 Hours	17%
	Need, Relevance and Significance of Values General, Concept and Meaning of Values and Ethics		
2.	Elements and Principles of Values	8 Hours	26%
	Universal & Personal Values, Social, Civic & Democratic Value		
3.	Applied Ethics	8 Hours	26%
	Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics		
4.	Value, Ethics & Global Issues	5 Hours	17%
	Cross-Cultural Issues, Role of Ethics & Values in Sustainability		
5.	Contemporary Issues in Values and Ethics	4 Hours	14%
	Case Studies, Presentations, Projects		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	1	3	2	1	-	-	-	-
CO2	-	-	-	-	-	1	-	-	1	-	-	2	-	2
CO3	-	-	-	-	-	3	1	-	-	-	-	-	-	1
CO4	-	-	-	-	-	3	1	1	-	-	-	-	-	-
CO5	-	-	1	1	-	-	-	-	2	-	-	-	-	1
CO6	-	-	-	-	-	-	-	3	-	2	-	-	-	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Reference book:**

- Human Values and Ethics in Workplace, United Nations Settlement Program, 2006.
(http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
- Ethics for Everyone, Arthur Dorbin, 2009.
(<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>) .
- Values and Ethics for 21st Century, BBVA.
(https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

❖ Web material:

- www.ethics.org

B. Tech. (Computer Engineering) Programme

SYLLABI (Semester – 5)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE354: OPERATING SYSTEM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	0	6	5
Marks	100	50	0	150	

Pre-requisite courses:

- Introduction to computer and computer organization.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	03
2.	Process Management	07
3.	Inter process Communication	09
4.	Deadlock	07
5.	Memory Management	12
6.	Input Output Management	04
7.	File Systems	06
8.	System Protection	04
9.	Unix/Linux File System	04
10.	Case study of XV6 operating system	04
	Total hours (Theory) :	60
	Total hours (Lab) :	30
	Total hours :	90

Detailed Syllabus:

1.	Introduction	03 Hours	05%
	What is an OS? Evolution Of OS, OS Services, Types Of OS, Concepts of OS, Different Views Of OS		
2.	Process Management	07 Hours	12%
	Process, Process Control Block, Process States, Threads, Types of Threads and Dispatching, Concurrent Threads, CPU Scheduling, Protection: Address space, Address Translation		
3.	Inter process Communication	09 Hours	15%
	Race Conditions, Critical Section, Co-operating Thread/ Mutual Exclusion. Hardware Solution, Strict Alternation, and		

	Peterson's Solution. The Producer Consumer Problem, Semaphores, Event Counters, Monitors. Message Passing and Classical IPC Problems: Reader's & Writer Problem, Dining Philosopher Problem.		
4.	Deadlock	07 Hours	12%
	Deadlock Problem, Deadlock Characterization, Deadlock Detection, Deadlock recovery, Deadlock avoidance: Banker's algorithm for single & multiple Resources. Deadlock Prevention.		
5.	Memory Management	12 Hours	20%
	Multiprogramming with Fixed and variable partitions, Paging: Principle Of Operation, Page Allocation, H/W Support For Paging. Multilevel paging, Segmentation, Swapping. Virtual Memory: Concept, Performance of Demand Paging, Page Replacement Algorithms, Thrashing and Working Sets.		
6.	Input Output Management	04 Hours	05%
	I/O Devices, Device Controllers, Direct Memory Access. Principles Of Input/output S/W : Goals Of The I/O S/W, Interrupt Handler, Device Driver, Device Independent. I/O Software Disks: RAID levels, Disk Scheduling Algorithm, Error Handling		
7.	File Systems	06 Hours	16%
	File Naming, File Structure, File Types, File Access, File Attributes, File Operations, Memory Mapped Files. Directories: Hierarchical Directory System, Pathnames, Directory Operations. File System Implementation, Contiguous Allocation, Linked List Allocation, Linked List Using Index, Inodes.		
8.	System Protection	04 Hours	06%
	Goals of protection, Principles of protection, Domain of protection, Access matrix, Implementation of Access matrix, Access control		
9.	Unix/Linux File System	04 Hours	04%
	Buffer Cache, Inodes, The system calls - ialloc, ifree, namei, alloc and free. Mounting and Unmounting, file systems, Network File systems. EXT file system in linux		
10.	Case study of XV6 operating system	04 Hours	05%

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Visualize and understand Operating system functionality and working of OS. Understanding of functionality, services of operating system and differentiate between different types of OS.
CO2	Define thread and process. Visualize how processes and threads are managed by the operating systems. Simulate and analyse various process scheduling algorithms. Explain and analyse different Inter process communication techniques
CO3	Describe deadlock and classify detection, recovery, prevention and avoidance algorithms. Test scenarios to report deadlock
CO4	Compare and Evaluate various memory management schemes. Simulate and analyse Memory management algorithms. Identify and describe the role of I/O devices.
CO5	Understand the file systems. Understand the secondary storage and simulate disk scheduling algorithm.
CO6	Understand the basic commands of Linux systems and protection mechanism.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	2	-	-	-	-	-	-	1	-	-
CO2	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO3	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO4	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO5	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO6	3	2	-	-	2	-	-	-	-	-	-	1	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Modern Operating Systems -By Andrew S. Tanenbaum, Third Edition PHI
2. Operating System Concepts Avi Silberschatz, Peter Baer Galvin, Greg Gagne, Ninth Edition, Wiley

❖ Reference book:

2. Operating Systems, D.M. Dhamdhare, TMH
3. Operating Systems Internals and Design Principles, William Stallings, Seventh Edition, Prentice Hall

4. Unix System Concepts & Applications, Sumitabha Das, TMH
5. Unix Shell Programming, Yashwant Kanitkar, BPB Publications

❖ **Web material:**

1. NPTEL course: <https://nptel.ac.in/courses/106/106/106106144/>
2. Simulator: <https://cpudemo.azurewebsites.net/#/>

❖ **Software:**

1. Simulator: OS Sim (OS Concepts Simulator)

CE355: DESIGN & ANALYSIS OF ALGORITHMS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	-	6	5
Marks	100	50	-	150	

Pre-requisite courses:

- Data Structure and Algorithms
- Programming language

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Algorithm Analysis	10
2.	Greedy Algorithm	08
3.	Dynamic Programming	10
4.	Divide and Conquer Algorithm	07
5.	Exploring Graphs	10
6.	String Matching and Introduction to NP-Completeness	08
7.	Approximation Algorithms	07
	Total hours (Theory) :	60
	Total hours (Lab) :	30
	Total hours :	90

Detailed Syllabus:

1.	Introduction	10 Hours	17%
	Fundamentals of algorithms, Performance Analysis, Primitive Operations, Time Complexity and Space Complexity, The efficiency of algorithm, average and worst case analysis, elementary operation, Asymptotic Notation, Analysing control statement, Analysing Algorithm using Barometer, Solving recurrence Equation, Sorting Algorithm: Selection, Insertion, Bubble Sort		
2.	Greedy Algorithm	08 Hours	13%
	Greedy: Characteristics, greedy functions, Problem solving:		

	Making change problem, The Knapsack Problem, Dijkstra's Shortest paths; Job Scheduling Problem, Disjoint sets, Minimum Spanning trees (Kruskal's algorithm, Prim's algorithm, Huffman coding		
3.	Dynamic Programming	10 Hours	17%
	Dynamic Programming: The Principle of Optimality, Problem Solving: Calculating the Binomial Coefficient, Making Change Problem, Assembly Line-Scheduling Knapsack Problem, Shortest Path Matrix Chain Multiplication, Longest Common Subsequence, All Pairs Shortest Path (Floyd-Warshall), Travelling Salesman Problem, Bellman Ford Algorithm.		
4.	Divide and Conquer Algorithm	07 Hours	12%
	Multiplying large Integers Problem, Binary Search Sorting (Merge Sort, Quick Sort), Matrix Multiplication, Exponential		
5.	Exploring Graphs	10 Hours	17%
	An Introduction, Undirected Graph, Directed Graph, Breath First Search, Depth First Search, Graph coloring problem, Applications of BFS & DFS, Backtracking –The Knapsack Problem; The Eight Queens problem, Branch and Bound –The Assignment Problem, The Knapsack Problem		
6.	String Matching and Introduction to NP-Completeness	08 Hours	13%
	The naïve string-matching algorithm, The Rabin-Karp algorithm, KMP Algorithm for Pattern Searching, Boyer-Moore string-search algorithm, The class P and NP Problem, Polynomial reduction, NP-Completeness Problem, NP-Hard problems		
7.	Approximation Algorithms	07 Hours	11%
	Vertex Cover Problem, Travelling Salesman Problem, Set Covering Problem, Randomization and Linear Programming		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Analyse the asymptotic performance of algorithms.
CO2	Derive time and space complexity of different sorting algorithms and compare them to choose application specific efficient algorithm.
CO3	Understand and analyse the problem to apply design technique from divide and conquer, dynamic programming, backtracking, branch and bound techniques and understand how the choice of algorithm design methods impact the performance of programs.
CO4	Understand and apply various graph algorithms for finding shortest path and minimum spanning tree.
CO5	Synthesize efficient algorithms in common engineering design situations.
CO6	Understand the notations of P, NP, NP-Complete and NP-Hard.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	-	-	-	-	-	-	-	-	1	-
CO2	2	2	-	-	-	-	-	-	-	-	-	2	2	-
CO3	3	3	3	3	2	-	-	-	-	-	-	2	2	-
CO4	2	3	3	1	-	-	-	-	-	-	-	-	2	-
CO5	1	-	1	-	-	-	-	-	-	-	-	2	1	1
CO6	3	1	-	-	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald Rivest and Clifford Stein, MIT Press

❖ **Reference book:**

1. Fundamental of Algorithms by Gills Brassard, Paul Bratley, Pentice Hall of India.
2. Fundamental of Computer Algorithms by Ellis Horowitz, Sartazsahni and sanguthevar Rajasekarm, Computer Sci.P.
3. Design & Analysis of Algorithms by P H Dave & H B Dave, Pearson Education.

❖ **Web material:**

1. <http://highered.mcgraw-hill.com/sites/0073523402/>

CE343: SOFTWARE ENGINEERING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- N/A

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Software and Software Engineering	04
2.	Agile Development	04
3.	Managing Software Project	05
4.	Requirement Analysis and Specification	04
5.	Software Design	05
6.	Software Coding & Testing	06
7.	Quality Assurance and Management	05
8.	Software Maintenance and Configuration Management	05
9.	Introduction to SaaS	03
10.	Advanced Topics in Software Engineering	04
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Introduction to Software and Software Engineering	04 Hours	09%
	The Evolving Role of Software, Software: A Crisis on the Horizon and Software Myths, Software Engineering: A Layered Technology, Software Process Models, The Linear Sequential Model, The Prototyping Model, The RAD Model, Evolutionary Process Models, Agile Process Model, Component-Based		

	Development, Process, Product and Process		
2.	Agile Development	04 Hours	09%
	Agility and Agile Process model, Extreme Programming, Other process models of Agile Development and Tools		
3.	Managing Software Project	05 Hours	11%
	Software Metrics (Process, Product and Project Metrics), Software Project Estimations, Software Project Planning (MS Project Tool), Project Scheduling & Tracking, Risk Analysis & Management(Risk Identification, Risk Projection, Risk Refinement ,Risk Mitigation)		
4.	Requirement Analysis and Specification	04 Hours	09%
	Understanding the Requirement, Requirement Modelling, Requirement Specification (SRS), Requirement Analysis and Requirement Elicitation, Requirement Engineering		
5.	Software Design	05 Hours	11%
	Design Concepts and Design Principal, Architectural Design, Component Level Design (Function Oriented Design, Object Oriented Design) (MS Visio Tool), User Interface Design, Web Application Design		
6.	Software Coding & Testing	06 Hours	13%
	Coding Standard and coding Guidelines, Code Review, Software Documentation, Testing Strategies, Testing Techniques and Test Case, Test Suites Design, Testing Conventional Applications, Testing Object Oriented Applications, Testing Web and Mobile Applications, Testing Tools (Win runner, Load runner)		
7.	Quality Assurance and Management	05 Hours	11%

	Quality Concepts and Software Quality Assurance, Software Reviews (Formal Technical Reviews), Software Reliability, The Quality Standards: ISO 9000, CMM, Six Sigma for SE, SQA Plan		
8.	Software Maintenance and Configuration Management	05 Hours	11%
	Types of Software Maintenance, Re-Engineering, Reverse Engineering, Forward Engineering, The SCM Process, Identification of Objects in the Software Configuration, Version Control and Change Control		
9.	Introduction to SaaS	03 Hours	07%
	Product Lifetime : Independent Product Vs. Continuous Improvement, Service Oriented Architecture, Cloud Computing, SaaS Architecture		
10.	Advanced Topics in Software Engineering	04 Hours	09%
	Component-Based Software Engineering, Client/Server Software Engineering, Web Engineering, Reengineering, Computer-Aided Software Engineering, Software Process Improvement, Emerging Trends in software Engineering		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand basics about software engineering principles, methods and practices and to analyze software requirement specification Prepare, SRS (Software Requirement Specification) document and SPMP (Software Project Management Plan) document.
CO2	Apply the concept of Functional Oriented and Object Oriented Approach for Software Design, To explain the software design strategies and to apply software measurement and metrics using Function point, Cyclomatic complexity and Halstead software science measures.
CO3	Recognize how to ensure the quality of software product, different quality standards and software review techniques.
CO4	Formulate problem by following Software Testing Life Cycle. Apply various testing techniques and test plan in. Design Manual Test cases for Software Project. Use

	automation testing tool students will be able test the software.
CO5	Able to understand modern Agile Development and Service Oriented Architecture Concept of Industry.
CO6	Analyze software risk with estimation parameters such as cost, effort, schedule/duration and understand the concepts of software maintenance, reengineering, reverse engineering, software configuration management.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	2	1	-	-	3	-	-	-	-	2	-
CO2	2	2	2	1	2	1	-	-	2	-	1	-	2	-
CO3	1	2	3	2	1	1	1	-	2	1	1	1	-	-
CO4	1	3	2	-	1	-	-	-	1	-	-	1	2	-
CO5	-	-	-	-	-	1	-	2	1	1	1	1	1	1
CO6	-	2	-	1	-	-	1	-	2	2	1	2	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Roger S. Pressman, Software engineering- A practitioner's Approach, McGraw-Hill International Editions

❖ Reference book:

1. Engineering Software as a Service An Agile Software Approach, Armando Fox and David Patterson
2. Ian Sommerville, Software engineering, Pearson education Asia
3. Pankaj Jalote, An Integrated Approach to Software Engineering by, Springer
4. Rajib Mall, Fundamentals of software Engineering, Prentice Hall of India.
5. John M Nicolas, Project Management for Business, Engineering and Technology, Elsevier

❖ **Web material:**

1. www.en.wikipedia.org/wiki/Software_engineering
2. www.win.tue.nl
3. www.rspa.com/spi
4. www.onesmartclick.com/engsineering/software-engineering.html
5. www.sei.cmu.edu
6. <https://www.edx.org/school/uc-berkeleyx>

❖ **Software:**

1. Wireshark
2. Cisco packet tracer
3. Network Simulator

CE360: MACHINE LEARNING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Probability, linear algebra, calculus and programming language

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Data cleaning and Preprocessing	04
2.	Supervised Machine Learning	18
3.	Unsupervised Machine Learning	09
4.	Reinforcement Learning	06
5.	Deep Neural Networks	08
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Data Cleaning and Pre-processing	04 Hours	07%
	Handling Missing data, Remove Noisy data, Feature selection, Scaling of features		
2.	Supervised Machine Learning	18 Hours	40%
	What is supervised ML, Linear Regression for univariate and Multivariate data, Cost function, Gradient Descent, Logistic regression, Under fitting and Over fitting, Support Vector Machine Kernels in SVM, Artificial Neural Network architecture, Activation functions, Forward pass in ANN, Back propogation in ANN, Model Evaluation techniques		
3.	Unsupervised Machine Learning	09 Hours	18%
	What is unsupervised ML, Different clustering algorithm,		

	Evaluate clustering algorithm		
4.	Reinforcement Learning	06 Hours	15%
	What is Reinforcement Learning?, RL task formulation (action space, state space, environment definition, reward), Q-Learning algorithm		
5.	Deep Neural Networks	08 Hours	20%
	Why Convolutional Neural Network, Different layers of CNN, padding, stride, CNN architecture and application in Computer vision, Transfer Learning, Why Recurrent Neural Network, Recurrent layer RNN application for NLP		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Apply appropriate data cleaning and data preprocessing techniques.
CO2	Create supervised ML mode.
CO3	Create unsupervised ML mode.
CO4	Design Reinforcement learning model.
CO5	Design advance ML model for Computer vision and Natural language processing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	1	-	-	-	-	-	-	-	-	-	-
CO2	3	-	3	2	-	-	-	-	-	-	-	-	1	-
CO3	3	-	3	2	-	-	-	-	-	-	-	-	1	-
CO4	3	-	3	2	-	-	-	-	-	-	-	-	1	-
CO5	3	-	3	2	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, 2nd Edition, Aurelien Géron, O'Reilly
2. Machine Learning Yearning, Andrew Ng, GitHub; eBook (Draft, 2018); eBook (MIT Licensed)
3. Reinforcement Learning: An Introduction, Richard S. Sutton and Andrew G. Barto, The MIT Press Cambridge, Massachusetts London, England
4. Deep Learning, Ian Goodfellow and Yoshua Bengio and Aaron Courville, The MIT Press (17 December 2016)

❖ Reference book:

1. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
2. Richard O. Duda, Peter E. Hart & David G. Stork, "Pattern Classification. Second Edition", Wiley & Sons, 2001.
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, "The elements of statistical learning", Springer, 2001.

❖ Web material:

1. <https://www.youtube.com/watch?v=foHSmB48rY&list=PLKvX2d3IUq586Ic9gIhZj6ubpWV-OJfl4>
2. https://www.youtube.com/watch?v=CS4cs9xVecg&list=PLkDaE6sCZn6EcXTbcX1uRg2_u4xOEky0
3. <https://www.youtube.com/watch?v=UzxYlbK2c7E>
4. <https://www.youtube.com/playlist?list=PLAwXTw4SYaPkQXg8TkVdIvYv4HfLG7SiH>
5. Recommended online course:
6. Coursera: <https://www.coursera.org/specializations/machine-learning-introduction>
7. Coursera: <https://www.coursera.org/specializations/deep-learning>

❖ Software:

1. Anaconda
2. GoogleColab

CE380: ADVANCED JAVA PROGRAMMING (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	4	-	6	4
Marks	100	100	-	200	

Pre-requisite courses:

- Object oriented Language – Java
- DBMS
- Web Programming

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Hibernate Introduction	02
2.	The Persistence Life Cycle	03
3.	Mapping with Annotations	05
4.	Hibernate Query Annotations	05
5.	Getting Started with Spring Boot	03
6.	Building REST APIs Using Boot	04
7.	Building Microservice using Spring Boot	08
	Total hours (Theory) :	30
	Total hours (Lab) :	60
	Total hours :	90

Detailed Syllabus:

1.	Hibernate Introduction	02 Hours	6%
	Origins of Hibernate and Object/Relational Mapping, Hibernate as a Persistence Solution, A Hibernate Hello World Example, The Steps Needed to Integrate and Configure Hibernate, Understanding Where Hibernate Fits into Your Java Application, Writing, reading and Updating data		

2.	The Persistence Life Cycle	03 Hours	10%
	Introducing the life cycle, Entities, Classes and Names, Identifiers, Entities and Associations, Saving Entities, Object Equality and Identity, 'Loading, Refreshing, Merging, Updating and Deleting Entities', Cascading Operations, Lazy Loading, Proxies, Collection Wrappers.		
3.	Mapping with Annotations	05 Hours	17%
	Primary keys, Lazy Loading, Associations (One-to-one, One-to-Many and Many-to-One, Many-to-Many), Specification of database (Column types and Sizes), Cons and Pros of annotation, JPA 2 Persistence Annotations (Entity Beans, Primary keys, Computing Primary Keys, inheritance, etc.), Configuring the annotated classes, Hibernate-specific Persistence Annotations (immutable, Natural IDs).		
4.	Hibernate Query Language	05 Hours	17%
	Syntax Basics (Update, Delete, Insert, Select), Logging and Commenting the Underlying SQL, Using restrictions with HQL, Using named parameters, Using Criteria API (Using restrictions with criteria, paging through the result set, obtaining a unique result, sorting the query's result), When to use filters, Defining and attaching Filters (Filters with annotations, Filter with XML Mapping Document).		
5.	Getting Started with Spring Boot	03 Hours	10%
	What is REST API, Connecting spring boot to the project, Connecting spring boot to the database, Creating user model, Creating Controller		
6.	Building REST APIs Using Spring Boot	04 Hours	13%
	Overview of Spring, Spring Architecture, bean life cycle, XML Configuration on Spring, Aspect – oriented Spring Managing Database, Managing Transaction		
7.	Building Microservice using Spring Boot	08 Hours	27%
	What are microservices, Why build Microservices, Microservices Architecture, Microservice resilience with Spring Cloud, Build streaming data microservices with Spring Cloud Stream		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Illustrate database access and details for managing information using the JDBC API
CO2	Design to map java classes and object associations to relational database tables with hibernate framework.
CO3	Demonstrate web application lifecycle using servlet, and JSP.
CO4	Identify the user interface control and build custom tag library using JSF
CO5	Develop real-world applications using java frameworks and understands about MVC architecture with its application.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	1	2	-	-	-	-	-	-	-	1	-
CO2	3	2	3	1	3	-	-	-	-	-	-	-	2	-
CO3	2	2	2	2	3	-	-	-	-	-	-	-	-	-
CO4	1	2	2	1	3	-	-	-	-	-	-	-	1	-
CO5	2	3	3	2	3	-	-	-	-	-	-	-	3	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. Joseph B. Ottinger, Jeff Linwood and Dave Minter Begging Hibernate, Apress Publications.
2. Beginning Spring Boot 3, K. Siva Prasad Reddy, Sai Upadhyayula, Apress Publications.
3. Spring Microservices in Action, John Carnell, Manning Books.

❖ **Reference book:**

1. JDBC™ API Tutorial and Reference, Third Edition, Maydene Fisher, Jon Ellis, Jonathan Bruce, Addison Wesley
2. Beginning JSP, JSF and Tomcat, Giulio Zambon, Apress
3. JSF2.0 CookBook, Anghel Leonard, PACKT publication

❖ **Web material:**

1. http://www.service-architecture.com/application-servers/articles/j2ee_web_site_architecture.html
2. <http://www.oracle.com/technetwork/java/javaee/overview/index.html>
3. <http://www.roseindia.net/struts/hibernate-spring/index.shtml>

❖ **Software:**

1. JDK 8+
2. NetBeans IDE
3. Eclipse
4. IntelliJ IDEA

CE381: MOBILE APPLICATION DEVELOPMENTS (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	4	-	6	4
Marks	100	100	-	200	

Pre-requisite courses:

- Basic Design concept with XML, Database management system.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Kotlin	02
2.	Functional Programming	03
3.	Objected oriented programming	05
4.	Dart Programming	06
5.	Flutter User Interface (Flutter UI)	06
6.	Theming, Styling and Routing	08
	Total hours (Theory) :	30
	Total hours (Lab) :	60
	Total hours :	90

Detailed Syllabus:

1.	Introduction to Kotlin	02 Hours	06%
	Introduction, Why Kotlin?, Data types, Functions, Looping & Ranges, Expression & statements, Objects Everywhere, Creating Class, Constructors, Packages, Exceptions, Property Accessors, overloading, Enumerations, Data Classes, Self Calls & the Elvis Operator, Introduction to Generics.		
2.	Functional Programming	03 Hours	10%
	Lambdas, The importance of Lambdas, Operations on Collections, Member References, Higher-Order Functions, Manipulating Lists, Building Maps, Sequences, Local Functions, Folding Lists,		

	Recursion		
3.	Objected oriented programming	05 Hours	17%
	Interfaces, Complex Constructors, Secondary Constructors, Inheritance, Base class initialization, Abstract Classes, Upcasting, Polymorphism, Composition, Inheritance & Extensions, Class Delegation, Down casting, Sealed Classes, Sealed Classes, Type Checking, Nested Classes, Objects, Inner Classes, Companion Objects		
4.	Dart Programming	06 Hours	20%
	Dart Introduction, Importing and using a library, Creating dart libraries, dart packages, Packages structure, Stagehand – The Dart project generator, The pubspec file, Package dependencies, Introducing async programming with features and isolates, Introducing unit testing with dart.		
5.	Flutter User Interface (Flutter UI)	06 Hours	20%
	Introduction to Flutter, Comparison with other mobile app development frameworks, flutter compilation & rendering, widgets introduction, hello flutter, Understanding built-in widgets, Creating UI with widgets, creating custom widgets, Handling user gestures, validating input, custom input and form field.		
6	Theming, Styling and Routing	08 Hours	27%
	Themes widgets, Material Design, iOS Cupertino, Using custom fonts, Dynamic styling with MediaQuery and LayoutBuilder, Understanding the navigator widget. Names routes, Screen transitions, Hero animations.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Knowledge of Kotlin programming language: Solid understanding of the Kotlin programming language.
CO2	Understanding of flutter framework: Through understanding of the flutter framework and its architecture.

CO3	Proficiency in developing mobile application: Developing mobile applications using the flutter and kotlin
CO4	Ethical consideration: Understand the ethical considerations involved in mobile application development

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	1	2	-	-	-	-	-	-	-	1	-
CO2	3	2	3	1	3	-	-	-	-	-	-	-	2	-
CO3	2	2	2	2	3	-	-	-	-	-	-	-	-	-
CO4	1	2	2	1	3	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Bruce Eckel and Svetlana Isakova, Atomic Kotlin, Mindview LLC.
2. Alessandro Biessek, Flutter for Beginners, Packt Publishing Limited

❖ Reference book:

1. "Android Programming: The Big Nerd Ranch Guide" by Bill Phillips, Chris Stewart, and Kristin Marsicano
2. "iOS Programming: The Big Nerd Ranch Guide" by Christian Keur and Aaron Hillegass
"Flutter for Beginners: An introductory guide to building cross-platform mobile applications with Flutter and Dart" by Alessandro Biessek

❖ Web material:

1. <https://kotlinlang.org/>
2. <https://flutter.dev/>

❖ **Software:**

1. Android Studio
2. Xcode
3. Visual Studio

CE382: ADVANCED WEB TECHNOLOGY(PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	4	0	6	4
Marks	30	70	0	100	

Pre-requisite courses:

- HTML, CSS, JavaScript

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	JavaScript Essentials	03
2.	JavaScript Modules and Object Oriented programming	02
3.	Asynchronous JavaScript	03
4.	Learning about the Internals of Node.js	03
5.	Working with ExpressJs and Form Handling	06
6.	Authentication, Authorization and Security	03
7.	MongoDB	05
8.	Sequelize ORM Framework	05
	Total hours (Theory) :	30
	Total hours (Lab) :	60
	Total hours :	90

Detailed Syllabus:

1.	JavaScript Essentials	03 Hours	10%
	Variables, Primitive data types, Analyzing and modifying data types, Operators, JavaScript Multiple Values, Arrays and their properties, Array methods, Multidimensional arrays, Objects in JavaScript, Working with objects and arrays, Basic functions, Parameters and arguments, Special functions and operators, Returning function values, Variable scope in functions, Recursive functions, Nested functions, Anonymous functions		
2.	JavaScript Modules and Object Oriented programming	02 Hours	07%
	Object-oriented programming, Classes and objects, Classes		

	Inheritance, Prototypes, Modules with Classes, Objects, and Closures, Modules in ES6: ES6 Exports, ES6 Imports, Imports and Exports with Renaming, Re-Exports, JavaScript Modules on the Web, Dynamic Imports with import(), Iterators and Generators: yield* and Recursive Generators, Advanced Generator Features		
3.	Asynchronous JavaScript	03 Hours	10%
	Asynchronous Programming with Callbacks, Promises, async and await, Asynchronous Iteration		
4.	Learning about the Internals of Node.js	03 Hours	10%
	NodeJs Architecture, Understanding event Loop, Package manager: npm and yarn, Create and publish custom package Understanding Requests, Sending Responses, Routing Requests, Redirecting Requests, Parsing Request Bodies Understanding Event Driven Code Execution, Blocking and Non-Blocking Code		
5	Working with ExpressJs and Form Handling	06 Hours	19%
	Adding Middleware, How Middleware Works? Express.js - Looking Behind the Scenes, Handling Different Routes, Parsing Incoming Requests, Limiting Middleware Execution to POST Requests, Using Express Router, Adding a 404 Error Page, Filtering Paths, Model View Controller Working with Forms, Validation and sanitization, Form Handling with ExpressJS, Using Fetch to Send Form Data, Sending Email and File Upload and Download		
6	Authentication, Authorization and Security	03 Hours	10%
	Externalizing Credentials, Cookies in Express, Examining Cookies, Sessions: Memory Stores, Using Sessions, Using Sessions to Implement Flash Messages, JSON Web Token(JWT), Rate Limiting		
7	MongoDB	05 Hours	17%
	Overview of MongoDB Fundamentals - the document model, how to create, update, read, and delete documents, indexing, and also learn about MongoDB Atlas, MongoDB databases, collections, and documents, Connecting to a MongoDB Database, MongoDB CRUD Operations: Replace and Delete Documents, Modifying Query		

	Results, MongoDB Aggregation, MongoDB Indexes, Data Modeling, Transaction.		
8	Sequelize ORM Framework	05 Hours	17%
	Defining and Using Sequelize Models: Defining models for a database, Exploring the various Sequelize data types and when to use them, Migrating schematic changes and data from Sequelize to the database, Manipulating and querying data using Sequelize, Validating, Customizing, and Associating Your Data, Validating Models, Associating Models, Implementing Transactions with Sequelize		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	How to use Express JS framework with Node.js with real projects with request (Understand)
CO2	Handle Form, session and cookies.
CO3	Render Client-side/Server-side website using template engine (Apply)
CO4	Create CRUD operations with MongoDB and Mongoose (Create)
CO5	Build a fast, scalable, feature-rich RESTful API (includes filters, sorts, pagination, and much more) (Create)
CO6	Build a complete, beautiful & real-world application from start to finish (API and server-side rendered website)

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	1	3	2	3	1	-	2	1	1	3	3	1	1
CO2	-	1	3	2	3	1	-	1	1	1	3	3	1	1
CO3	-	1	3	2	3	1	-	1	2	2	3	3	2	1
CO4	-	1	3	2	3	2	-	1	2	2	3	3	3	1
CO5	-	1	3	2	3	2	-	1	2	3	3	3	3	1
CO6	-	1	3	2	3	3	-	2	3	3	3	3	3	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Web Development with Node and Express, 2nd Edition by Ethan Brown, O'Reilly Media

❖ **Reference book:**

1. Node.js Web Development - Fifth Edition By David Herron, Packtpub
2. Node Cookbook - Fourth Edition By Bethany Griggs, Packtpub

❖ **Web material:**

1. <https://nodejs.org/en/docs/>
2. <https://handlebarsjs.com/guide/>

❖ **Software:**

1. Node.js, Express.js, MondoDB
2. IDE: Visual code Editor, Atom, Sublime

CE356: SOFTWARE GROUP PROJECT-III

Credits and Hours:

Teaching Sch	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	4	-	4	2
Marks	-	100	-	100	

Pre-requisite courses:

- Programming Language, Software Engineering.

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - Project Synopsis
 - Software Requirement Specification
 - SPMP
 - Final Project Report
 - Project Setup file with Source code
 - Project Presentation (PPT)

- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 30

Total hours : 30

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	An ability to function effectively in teams to accomplish a common goal.
CO2	An ability to apply knowledge of computing and engineering to evaluate project requirements.
CO3	An ability to design, implement and evaluate a computer-based system, process, component, or program to meet desired needs.
CO4	An ability to analyze the local and global impact of computing on individuals, organizations, and society.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	An ability to communicate effectively with end user.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	2	-	1	2	-
CO2	3	-	1	2	-	-	-	-	-	-	-	-	3	-
CO3	-	2	3	-	1	-	-	-	-	-	-	-	2	-
CO4	-	-	-	-	-	3	2	-	-	1	-	-	2	-
CO5	-	-	-	-	-	-	1	-	-	2	-	3	-	2
CO6	-	-	-	-	-	-	3	1	-	-	-	2	-	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Books, Magazines ,Journals & online course platforms of related topics

❖ Web material:

1. www.sciencedirect.com
2. www.elsevier.com
3. <https://www.udemy.com/>
4. <https://www.udacity.com/>
5. <https://nptel.ac.in/course.html>
6. <https://www.futurelearn.com/>

❖ Software:

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS

CE346: SUMMER INTERNSHIP-I

Credits and Hours:

Teaching Scheme	Project	Practical	Tutorial	Total	Credit
Hours	90	-	-	90	3
Marks	150	-	-	150	

Objectives of the Course:

- To get familiar with modern tools and technologies use in company/industry/organization
- To get involved in design, development and testing practices followed in the company/industry/organization
- To enhance their soft-skills, presentation skills, interpersonal skills, documentation skills and office etiquettes required to sustain in company/industry/organization environment
- To participate in teamwork and preferably as part of a multi-disciplinary team
- To make them aware about company/industry/organization best practices, processes and regulations.
- To make them more productive, consistent and punctual.

Outline of the Course:

1.	Instructional Method and Pedagogy
	• Summer internship shall be at least 90 hours during the summer vacation only. • Department/Institute will help students to find an appropriate company/industry/organization for the summer internship. • The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline. • Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed. • During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and those of the University. • Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.

	• The student shall submit two documents to the Coordinator for the evaluation of the summer internship: • Summer Internship Report • Summer Internship Assessment Form • Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted through the presentation to the Coordinator by the first day of the new term. • The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester. • If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.
2.	Format of Summer Internship Report
	The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections: • Cover Page • Page of Approval and Grading • Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at. • TABLE OF CONTENTS (with the corresponding page numbers) • LIST OF FIGURES AND TABLES (with the corresponding page numbers)

- **DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION:**
Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- **INTRODUCTION:** In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- **PROBLEM STATEMENT:** What is the problem you are solving, and what are the reasons and causes of this problem.
- **SOLUTION:** In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
- **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it

	and waste tons of paper.	
	Total hours (Project) :	90
	Total hours :	90

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Ability to integrate existing and new technical knowledge for industrial application.
CO2	Executing work with team and teammates from other disciplines
CO3	Get practices and experience related to professional and ethical issues in the work environment
CO4	Experience of demonstrating the impact of the internship on their learning and professional development.
CO5	Understanding of lifelong learning processes through critical reflection of internship experiences.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	2	1	-	-	2
CO3	-	-	-	-	-	-	1	3	-	1	-	-	-	1
CO4	-	-	-	-	-		3	1	-	-	-	-	1	-
CO5	-		-	-	-	1	-	-	-	-	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Books, Magazines & Journals of related topics

❖ **Web material:**

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

❖ **Software**

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS
5. FLUTTER
6. NODE/REACT NATIVE

HS 131.02 A: COMMUNICATION AND SOFT SKILLS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communicative English

Objectives of the Course:

- To hone and sharpen Communication Skills of students
- To prepare globally and multi-culturally competent communicators and professionally compatible cadre of future professionals
- To equip and empower students to qualify and successfully clear all the phases of selection procedure for on and off campus interviews

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	An Introduction to Communication	06 Hours	20%
	Basics of Communication: Origin, Concept, Process, Levels, Principles and Barriers; Applications of Communication; Rhetoric in Professional Communication; Importance of Ethos, Logos, and Pathos in Communication		

2.	Cross-cultural Communication and Globalization	03 Hours	10%
	Basic Concepts: Culture, Globalization and Cross-cultural Communication; Social and People Skills; Communicating with People of Different Cultures; Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them; Persuasive Communication		
3.	Communication for Career Building	10 Hours	33%
	Cover Letters and Resume; E-mail and Report; Types of Resume; Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion; Concept and Rationale of Personal Interview; Types of Personal Interview; Writing Statement of Purpose		
4.	Group Dynamics and Soft Skills	05 Hours	17%
	An Introduction to Group Dynamics and Soft Skills; Groups and their Structures; Roles and Functions of Members in Groups; Conflict Management; Aptitude and Attitude; Various Intelligences; Developing an Open Mindset		
5.	Effective Presentation Strategies	04 Hours	14%
	Designing Appealing Presentation; Audience Analysis and Supporting Material; Presentation Mechanics and Presentation Process; Managing Yourself during Q and A Session; Fundamentals of Persuasion		
6.	Contemporary Issues in Communication and Soft Skills	02 Hours	06%
	Trends and Practices in Communication, Case Studies		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Gain thorough understanding and proficiency in various Professional Communication Skills.
CO2	Develop awareness and competence in cross-cultural communication in their personal, academic and professional environments.
CO3	Develop business writing and presentation skills to succeed in career.
CO4	Develop soft skills to stand out and take their career to the next level.
CO5	Develop various intelligences and open Mindset to function in multi-disciplinary and cross-cultural work environment.
CO6	Practice new trends in communication in multiple perspectives at personal,

professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	2	2	2	-	2	-	-	-	2
CO2	-	-	2	-	-	3	-	-	3	-	-	-	-	3
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	1	-	-	2	1	3	2	-	-	-	2
CO5	-	-	-	-	-	2	2	-	-	-	2	-	-	2
CO6	-	-	-	-	-	-	-	-	-	3	-	2	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Koneru, A. Professional Communication, Tata McGraw Hill Education Private Limited
2. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
3. Raman, M & Singh, P. Business Communication, Oxford University Press

❖ **Reference book:**

1. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
2. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGraw Hill Education Private Limited

❖ **Web material:**

1. <https://www.coursera.org/learn/careerdevelopment>
2. <https://www.futurelearn.com/courses/writing-applications>
3. <https://www.futurelearn.com/courses/workplace-englis>

B. Tech. (Computer Engineering) Programme

SYLLABI (Semester – 6)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE357: ARTIFICIAL INTELLIGENCE

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Pre-requisite courses:

- Linear algebra (vectors, matrices, derivatives)
- Calculus.
- Basic probability theory.
- Python programming.

Outline of the course:

Sr No.	Title of the unit	Minimum number of hours
1.	Fundamental concepts	04
2.	Heuristic Search Techniques Problems, Problems Space and Search, Heuristic Search Techniques	08
3.	Knowledge Representation and Logic	05
4.	Reasoning	06
5.	Weak Slot-And-Filler Structure, Game Playing and Planning	06
6.	Natural Language Processing (NLP), Text Analytics and Neural Networks	08
7.	Advanced Topics	08
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1	Fundamental concepts	04 hours	10%
	The History of Artificial Intelligence, The AI Problems, AI Techniques, Applications of AI, Strong AI vs Weak AI, The Turing Test, Agents and Environments, The concept of Rationality, The Nature of Environments and the Structure of Agents.		
2	Problems, Problems Space and Search, Heuristic Search Techniques	08 hours	18%

	Defining the Problems as a State Space Search, Production Systems, Problem Characteristics, Production System Characteristics, Issues in The Design of Search Programs, Generate-And-Test, Hill Climbing, Simulated Annealing, Best-First Search, A*, AO*, Branch and Bound Problem Reduction, Constraint Satisfaction, Means-Ends Analysis.		
3	Knowledge Representation and Logic	05 hours	12%
	Knowledge Representations and Mappings, Approaches to Knowledge Representation., Issues in Knowledge Representation, Representing Knowledge using Rules: Procedural Versus Declarative Knowledge, Logic Programming, Forward Versus Backward Reasoning, Matching, Control Knowledge. Logic: Instance and ISA Relationship, Computable Functions and Predicates, Resolution.		
4	Reasoning	06 hours	12%
	Symbolic Reasoning Under Uncertainty and Statistical Reasoning: Introduction to Non-monotonic Reasoning. Statistical Reasoning: Probability and Bayes' Theorem, Certainty Factors and Rule-Based Systems, Bayesian Networks, Dempster-Shafer Theory, Fuzzy Logic.		
5	Weak Slot-And-Filler Structure, Game Playing and Planning	06 hours	12%
	Weak Slot-And-Filler Structure: Semantic Nets, Frames. Game Playing: The MiniMax Search Procedure, Adding Alpha-Beta Cutoffs, Additional Refinements, Iterative, Deepening. Planning: Blocks World, STRIPS, Constraint Satisfaction, Basics of Probabilistic Planning.		
6	Natural Language Processing (NLP), Text Analytics and Neural Networks	08 hours	18%
	NLP and Text Analytics: Introduction, Syntactic Processing, Semantic Analysis, Semantic Analysis, Discourse and Pragmatic Processing, Text Analytics, Text pre-processing, Bag of Words, Word Cloud, Machine Translation, sentiment analysis. Neural Networks: Introduction: Simple Perceptron, Hopfield Network, Learning in Neural Network, Application of Neural Networks, Recurrent Networks, Deep Neural Network, Convolution Network, Restricted Boltzmann machine, Transfer learning		
7	Advanced Topics	08 hours	18%

	Optimization Techniques: Genetic Algorithm (GA), Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), Tabu Search, Introduction to topics like Computer Vision, Expert Systems, Big data, Neuro Computing, Robotics, Web Search.		
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Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the various searching techniques, constraint satisfaction problem and example problems- game playing techniques.
CO2	Apply these techniques in applications, which involve perception, reasoning and learning.
CO3	Demonstrate fundamental understanding of the history of artificial intelligence (AI) and its foundations.
CO4	Demonstrate awareness and a fundamental understanding of various applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.
CO5	Demonstrate proficiency-developing applications in an 'AI language', expert system shell, or data-mining tool.
CO6	Demonstrate an ability to share in discussions of AI, its current scope and limitations, and societal implications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	1	1	1	1	1	1	-	-	2	2	2	3
CO2	2	2	-	-	-	-	-	1	2	1	1	1	2	-
CO3	3	1	1	-	-	1	1	-	-	-	-	-	2	-
CO4	3	2	3	3	-	-	-	1	2	2	2	2	3	3
CO5	3	2	2	2	3	2	2	-	-	-	-	-	2	-
CO6	2	1	1	-	-	3	-	-	-	-	2	2	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text Books:

1. “Artificial Intelligence” -By Elaine Rich and Kevin Knight (2nd Edition) Tata Mcgraw-Hill.
2. Stuart Russell, Peter Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009. Can also use 2nd Ed., Pearson Education International, 2003.

3. Introduction to Prolog Programming by Carl Townsend.

❖ **Reference Books:**

1. Artificial Intelligence and Expert System by D.W. Patterson, PHI
2. Introduction to Applied Fuzzy Logic by Ahmed Abraham, PHI
3. Nils Nilsson, Artificial Intelligence: A New Synthesis, Morgan Kaufmann, 1998.
4. David Poole, Alan Mackworth, Artificial Intelligence: Foundations for Computational Agents, Cambridge Univ. Press, 2010.
5. Artificial Intelligence & design of expert systems, Lager, Benjamin/Cummings.
6. Artificial Intelligence - An Engineering Approach, Schalkoff R. J., McGraw-Hill.
7. Expert Systems: Theory & Practice, Jean-Louis Ermine, Prentice-Hall India, 1997.
8. Programming with PROLOG” –By Klocks in and Mellish.

❖ **Web Material:**

1. <http://stanford.edu/~cpiech/cs221/>
2. <https://www.expertsystem.com/examples-natural-language-processing-systems-artificial-intelligence/>
3. <https://www.linkedin.com/pulse/sentiment-analysis-cognitive-computing-jaish-mathews>
4. <https://www.johanahlen.info/en/2017/04/text-analytics-and-sentiment-analysis-with-microsoft-cognitive-services/>

❖ **Software:**

1. Python
2. SWI Prolog

CE348: INFORMATION SECURITY

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	-	6	5
Marks	100	50	-	150	

Pre-requisite courses:

- N/A

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	02
2.	Traditional Symmetric-Key Ciphers	05
3.	Introduction to Modern Symmetric-Key Ciphers	04
4.	Data Encryption Standard (DES)	04
5.	Advanced Encryption Standard (AES)	04
6.	Encipherment Using Modern Symmetric-Key Ciphers	04
7.	Asymmetric-Key Cryptography	05
8.	Message Integrity and Message Authentication	04
9.	Cryptographic Hash Functions	04
10.	Digital Signature	04
11.	Entity Authentication	04
12.	Key Management	04
13.	Security at the Application Layer: PGP and S/MIME	04
14.	Security at the Transport Layer: SSL and TLS	04
15.	Security at the Network Layer: IPSec	04
	Total hours (Theory) :	60
	Total hours (Lab) :	30
	Total hours :	90

Detailed Syllabus:

1.	Introduction	02 Hours	06%
	Security Goals, Attacks, Services and Mechanism and Techniques.		
2.	Traditional Symmetric-Key Ciphers	05 Hours	10%
	Introduction, Substitution Cipher, Transposition Cipher, Stream and Block Cipher		
3.	Introduction to Modern Symmetric-Key Ciphers	04 Hours	06%
	Modern Block Cipher, Modern Stream Cipher		
4.	Data Encryption Standard (DES)	04 Hours	05%
	Introduction, DES structure and Analysis, Multiple DES		
5.	Advanced Encryption Standard (AES)	04 Hours	05%
	Introduction, AES structure and Analysis		
6.	Encipherment Using Modern Symmetric-Key Ciphers	04 Hours	06%
	Use of Modern Block Ciphers, Use of Modern Stream Ciphers		
7.	Asymmetric-Key Cryptography	05 Hours	10%
	Introduction, RSA Cryptosystem, RABIN Cryptosystem, ELGAMAL Cryptosystem		
8.	Message Integrity and Message Authentication	04 Hours	08%
	Message Integrity, Message Authentication		
9.	Cryptographic Hash Functions	04 Hours	08%
	Introduction, SHA-512,MD5		
10.	Digital Signature	04 Hours	06%
	Services,RSA Digital Signature Scheme,ELGamal Digital Signature Scheme		
11.	Entity Authentication	04 Hours	04%
	Passwords, Challenge –Response,Zero –Knowledge		
12.	Key Management	04 Hours	08%
	Symmetric-Key Distribution,KERBEROS		
13.	Security at the Application Layer: PGP and S/MIME	04 Hours	06%
	E-Mail, PGP,S/MIME		
14.	Security at the Transport Layer: SSL and TLS	04 Hours	06%

	SSL Architecture,TLS cipher suite		
15.	Security at the Network Layer: IPSec	04 Hours	06%
	Introduction to IPSec		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Define various security goal and understand the security policies such as the CIA triad of Confidentiality, Integrity and Availability.
CO2	Classify various forms of security attacks, where they arise, and appropriate tools or mechanism to quantify them.
CO3	Illustrate a basic understanding of cryptography, how it has evolved, and evaluate symmetric key encryption techniques used today.
CO4	Distinguish modern symmetric encryption standard, key distribution scenario and analyse effectiveness in today's environment.
CO5	Evaluate Asymmetric key encryption techniques, key distribution scenario and calculate public and private components of asymmetric key encryption techniques.
CO6	Develop message integrity and message authentication of message digest.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	-	-	3	1	-	2	1	-	-	-	-	-
CO2	2	3	3	2	3	1	-	-	-	-	-	-	1	-
CO3	2	2	3	2	3	2	-	-	2	1	-	-	2	-
CO4	2	2	3	2	3	-	2	-	2	1	-	-	3	1
CO5	3	2	3	2	1	-	2	1	1	-	-	-	3	1
CO6	2	2	3	1	-	-	2	2	1	-	2	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Cryptography and Network Security, Behrouz A. Forouzan, McGraw-Hill Companies.

❖ Reference book:

1. Cryptography And Network Principles And Practice, William Stallings, Prentice Hall, Pearson Education Asia.
2. Cryptography & Network Security, Atul Kahate, The McGraw-Hill Companies.
3. The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations by Gene Kim, Jez Humble, Patrick Debois, and John Willis

❖ Web material:

1. <http://people.csail.mit.edu/rivest/crypto-security.html>.
2. <http://www.cryptix.org/>
3. <http://www.cryptocd.org/>
4. <http://www.cryptopp.com/>
5. <http://www.freetechbooks.com/information-security-f52.html>

❖ Software:

1. Nmap
2. Wireshark

CE349: THEORY OF COMPUTATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	-	-	3	3
Marks	100	-	-	100	

Pre-requisite courses:

- Mathematics, Data Structures and Algorithms, Design and Analysis of Algorithm (Computational Complexity)

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction	03
2	Mathematical Terms and Theory	05
3	Regular Grammar & Languages, Regular Expression, Finite Automata	14
4	Context Free Grammar & Languages, Push down	13
5	Turing Machine, Recursively Enumerable Languages	08
6	Decidable & Undecidable Problems	02
	Total hours (Theory) :	45
	Total hours (Lab) :	00
	Total hours :	45

Detailed Syllabus:

1.	Introduction	03 Hours	06 %
	Alphabet, String, Language, Formal Grammar, Chomsky Hierarchy, Introduction to Automata		
2.	Mathematical Terms and Theory	05 Hours	10 %
	Mathematical Inductions, Recursive Definitions		
3.	Regular Grammar & Languages, Regular Expression, Finite Automata	14 Hours	32 %

	Regular Language, Regular Expressions, Applications, Chomsky Hierarchy, Finite Automata, Nondeterministic Finite Automata, Kleen's Theorem, Automata with Output (Moore Machine, Mealy Machine), Properties of Regular Languages (Pumping Lemma, Closure Property, Decision Algorithm)		
4.	Context Free Grammar & Languages, Push down Automata	13 Hours	30%
	The Chomsky, Notion of Grammars and Languages Generated by Grammars, CFG, CFL, Regular Language and Regular Grammar, Derivation Tree and Ambiguity, BNF, CNF, GNF, CFL properties (Pumping Lemma, Closure Property, Decision Algorithm), Intersections and Complements of CFL, Non-CFL, Definition, DPDA, NPDA, Equivalence of CFG and PDA		
5.	Turing Machine, Recursively Enumerable Languages	08 Hours	18%
	Definition, Model of Computation, Combining TM, Variations of TM, Non Deterministic TM, Universal TM, Recursively Enumerable and Recursive, Enumerable Languages, Context sensitive languages		
6.	Decidable & Undecidable Problems	02 Hours	04 %
	Tractable and Intractable Problems, Complexity Classes, Tractable and Possibly Intractable Problems, P and NP Completeness, Countability		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Apply basic concepts of theory of computation in the computer field in order to solve computational problems.
CO2	Construct algorithms for different problems and argue formally about correctness on different restricted machine models of computation.

CO3	Analyze and design finite automata, pushdown automata and Turing machine for formal languages.
CO4	Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.
CO5	Identify limitations of some computational models and possible solutions.
CO6	Design context free grammars for formal languages.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	-	3	-
CO2	-	-	3	-	-	-	-	-	-	-	-	-	2	-
CO3	3	2	2	-	-	-	-	-	-	-	-	-	3	-
CO4	2	2	2	-	-	-	-	-	-	-	-	-	1	-
CO5	2	1	1	-	-	-	-	-	-	-	-	-	-	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Introduction to Languages and Theory of Computation, John C. Martin, TMH

❖ Reference book:

1. An introduction to automata theory and formal languages, Adesh K. Pandey, S. K. Kataria & Sons
2. Introduction to computer theory, Deniel I. Cohen, John Wiley & Sons Inc
3. Computation: Finite and Infinite, Marvin L. Minsky, Prentice-Hall
4. “An introduction to Formal Languages and Automata”, Peter Linz, 6th edition, Jones & Bartlett Learning

5. “Introduction to the Theory of Computation”, Michael Sipser, 3rd edition, Cengage Learning.

❖ **Web material:**

1. https://www.youtube.com/playlist?list=PLEbnTDJUr_IdM___FmDFBz0zCsOFxfK
2. <http://nptel.ac.in/courses/106103070/>
3. <http://nptel.ac.in/courses/106104028/>
4. <http://nptel.ac.in/courses/106106049/>
5. <https://www.youtube.com/watch?v=4GLC-s0PQLY>

CE358: COMPUTER NETWORKS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Data Communication and Networking

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Computer Networks and the Internet	04
2.	Application Layer	07
3.	Transport Layer	17
4.	The Network Layer	08
5.	The Link Layer and Local Area Networks	07
6.	Current Trends in computer Network	02
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Computer Networks and the Internet	04 Hours	08%
	What Is a Protocol, Access Networks, Physical Media, Packet Switching & Circuit Switching, Delay, Loss, and Throughput in Packet-Switched Networks-Overview, Queuing Delay and Packet Loss, End-to-End Delay, Throughput in Computer Networks, Primer on Latency and Bandwidth		
2.	Application Layer	07 Hours	16%
	Principles of Network Applications, The Web and HTTP, File Transfer: FTP, SMTP, DNS, Optimizing Application Delivery		
3.	Transport Layer	17 Hours	38%
	Introduction and Transport-Layer Services, Multiplexing and Demultiplexing, Connectionless Transport: UDP & Building Blocks		

	of UDP, Principles of Reliable Data Transfer, Connection-Oriented Transport: TCP, Principles of Congestion Control & Building Blocks of TCP		
4.	The Network Layer	08 Hours	18%
	Introduction, Virtual Circuit and Datagram Networks, What's Inside a Router, The Internet Protocol (IP): Forwarding and Addressing in the Internet, Routing Algorithm(SS)		
5.	The Link Layer and Local Area Networks	07 Hours	16%
	Introduction to the Link Layer, Multiple Access Protocols, Link Layer Addressing(SS), Ethernet(SS)		
6.	Current Trends in computer Network	02 Hours	04%
	Limitations of TCP/IP protocol, Need of SDN, SDN architecture, Functionality of different SDN plane, Current trends*.		

* Teacher will decide content of current trends and communicate to students well in advance.

- **Course Outcome (COs):**

At the end of the course, the students will be able to

CO1	Measure and Analyse different network parameters.
CO2	Understand working of application layer protocols.
CO3	Understand and Analyse transport layer services and its impacts on data rate.
CO4	Understand basic functionality of network layer devices.
CO5	Understand functionality of multiple access protocol.
CO6	Analyse traditional network and get familiar with Software defined networking.

- **Course Articulation Matrix:**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	3	1	-	-	-	-	-	-	-	2	-
CO2	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO3	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO4	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	3	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text Book**

1. Computer Networking: A Top-Down Approach James F. Kurose, University of Massachusetts, Amherst Keith W. Ross, Polytechnic University, Brooklyn

❖ **Reference Materials**

1. Computer Networks by Andrew S Tanenbaum.
2. High Performance Browser Networking by Ilya Grigorik
3. Data Communication and Networking by Behrouz Forouzan

❖ **Web Materials**

1. www.ietf.org – For drafts
2. www.ieee.org – For standards and technical research papers
3. <http://nptel.iitm.ac.in/courses.php?disciplineId=117>

CE374: Service Oriented Computing (PE-II)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- N/A

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Overview of SOA	03
2.	SOA — Architecture Fundamentalsg	06
3.	Web Services and Contemporary SOA (Part I: Activity Management and Composition)	06
4.	Web Services and Contemporary SOA (Part II:	08
5.	Building SOA (Planning and Analysis)	06
6.	Building SOA (Technology and Design)	10
7.	Fundamental WS-* Extensions	06
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Overview of SOA	03 Hours	07%
	The Promise of SOA, The Challenges of SOA: Reuse, Efficiency in Development, Integration of Applications and Data, Agility, Flexibility, and Alignment.Meeting the Challenge Reference Architecture Common Semantics Governance Business Process Modelling Design-Time Service Discovery Model-Based Development		

2.	SOA — Architecture Fundamentals	06 Hours	13%
	Architectural Styles, Architectural Principles and Practices, SOA and Other Architectures, Enterprise Architecture, Software Architecture, EA, 4+1, and Services, What Is a Service?, Service Characteristics: Service, Granularity, Service Dimensions, Loose Coupling Is King: Location Transparency, Interface and Implementation, Data, Versioning, Interoperability and Platform Independence, Usage, Assumptions, and Knowledge, Common Service Patterns, Service Types and Purpose, SOA Reference Architecture.		
3.	Web Services and Contemporary SOA (Part I: Activity Management and Composition)	06 Hours	13%
	Message exchange patterns, Service activity, Coordination, Atomic transactions, Business activities, Orchestration		
4.	Web Services and Contemporary SOA (Part II:Advanced Messaging, Metadata, and Security)	08 Hours	18%
	Addressing, Reliable messaging, Correlation, Policies, Metadata exchange, Security, Notification and eventing		
5.	Building SOA (Planning and Analysis)	06 Hours	13%
	SOA Delivery Strategies: SOA delivery lifecycle phases, The top-down strategy, The bottom-up strategy, Service Modeling: Service modeling guidelines, Classifying service model logic.		
6.	Building SOA (Technology and Design)	10 Hours	23%
	WSDL-related XML Schema language basics, WSDL language basics, SOAP language basics, Composition Guidelines: Steps to composing SOA, Considerations for choosing service layers, Considerations for positioning core SOA standards Service Design: Service design overview, Entity-centric business service design (a step-by-step process), Application service design, Task-centric business service design Business Process Design: WS-BPEL language basics, WS-Coordination overview.		

7.	Fundamental WS-* Extensions	06 Hours	13%
	WS-Addressing, WS-ReliableMessaging, WS-Policy, WS-MetadataExchange, WS-SecuritySOA Platforms: SOA support in J2EE, SOA support in.NET.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Define various promise and challenges of service oriented architecture.
CO2	Classify various forms of architectures, integration and interoperability of architecture principles.
CO3	Illustrate a basic understanding of activity management, how it has evolved, and evaluate composition and message exchanging.
CO4	Classifying delivery life cycle phase and analyse service model logic.
CO5	Evaluate web services using SOAP, WSDL and create application service design.
CO6	Explore platforms such as J2EE and .Net for reliable messaging, security and policies for efficient service discovery.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	-	-	3	1	-	2	1	-	-	-	-	-
CO2	2	3	3	2	3	1	-	-	-	-	-	-	1	-
CO3	2	2	3	2	3	2	-	-	2	1	-	-	2	-
CO4	2	2	3	2	3	-	2	-	2	1	-	-	3	1
CO5	3	2	3	2	1	-	2	1	1	-	-	-	3	1
CO6	2	2	3	1	-	-	2	2	1	-	2	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Mike Rosen, Boris Lublinsky, Kevin T. Smith, Marc J. Balcer "Applied SOA:Service-Oriented Architectureand Design Strategies" WileyPublishing, Inc, 2008
2. Thomas Erl, “Service-Oriented Architecture: Concepts, Technology, and Design, Prentice Hall Publication, 2008

❖ **Reference book:**

1. Norbert Bieberstein, Sanjay Bose, Marc Fiammante, Keith Jones, Rawn Shah, “Service-Oriented Architecture Compass: Business Value, Planning, andEnterprise Roadmap”, IBM Press Publication, 2005.
2. Thomas Erl, “Service-Oriented Architecture: A Field Guide to Integrating XML and Web Services”, Prentice Hall Publication, 2004.
3. SanjivaWeerawarana, Francisco Curbera, Frank Leymann, Tony Storey, DonaldF.Ferguson, “Web Services Platform Architecture: SOAP, WSDL, WS-Policy, WS- Addressing, WS-BPEL, WS-Reliable Messaging, and More”, Prentice HallPublication, 2005.
4. Eric Newcomer, Greg Lomow, “Understanding SOA with Web Services”,Addison Wesley Publication, 2004.
5. ERL, T.; Service-oriented architecture : a field guide to integrating XML and Web services, Prentice Hall PTR, 2004, ISBN- 0131428985.

❖ **Web material:**

1. <https://www.ibm.com/developerworks/library/ws-top10/>
2. <http://www.oracle.com/technetwork/articles/javase/soa-142870.html>
3. <https://docs.oracle.com/javaee/6/tutorial/doc/gijti.html>.

CE378: DISTRIBUTED SYSTEMS(PE-II)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	4
Marks	100	50	0	150	

Pre-requisite courses:

- Introduction to computer and computer organization.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	03
2.	Distributed and Parallel Database Design	05
3.	Distributed Data Control	04
4.	Distributed Query Processing	07
5.	Distributed Transaction Processing	05
6.	Data Replication and Integration	06
7.	Parallel Database Systems	05
8.	Big Data Processing	05
9.	NoSQL, NewSQL, and Polystores	05
	Total hours (Theory):	45
	Total hours (Lab):	30
	Total hours:	75

Detailed Syllabus:

1.	Introduction	03 Hours	05%
	What Is a Distributed Database System? Promises of Distributed DBMSs, Distributed DBMS Architectures		
2.	Distributed and Parallel Database Design	05 Hours	10%
	Data Fragmentation, Allocation, Combined Approaches, Adaptive Approaches, Data Directory		
3.	Distributed Data Control	04 Hours	08%
	View Management, Access Control, Centralized Semantic Integrity Control, Distributed Semantic Integrity Control		
4.	Distributed Query Processing	07 Hours	18%
	Introduction, Data Localization, Join Ordering in Distributed		

	Queries, Distributed Cost Model, Distributed Query Optimization, Adaptive Query Processing		
5.	Distributed Transaction Processing	05 Hours	12%
	Background and Terminology, Distributed Concurrency Control, Distributed Concurrency Control Using Snapshot Isolation, Distributed DBMS Reliability, Modern Approaches to Scaling Out Transaction Management		
6.	Data Replication and Integration	06 Hours	15%
	Consistency of Replicated Databases, Update Management Strategies, Replication Protocols, Group Communication, Replication and Failures, Database Integration, Multidatabase Query Processing		
7.	Parallel Database Systems	05 Hours	14%
	Parallel Architectures, Data Placement, Parallel Query Processing, Load Balancing, Fault-Tolerance, Database Clusters		
8.	Big Data Processing	05 Hours	08%
	Distributed Storage Systems, Big Data Processing Frameworks, Data Lakes		
9.	NoSQL, NewSQL, and Polystores	05 Hours	10%
	Motivations for NoSQL, Key-Value Stores, Document Stores, Graph DBMSs, Polystores		

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand the concepts and issues related to distributed systems and parallel database.
CO2	Design and develop the application for distributed environment.
CO3	Manage performance, reliability and other issues while designing in distributed environment.
CO4	Apply knowledge of distributed systems techniques and methodologies.
CO5	Evaluate the performance of distributed systems.
CO6	Perform big data processing for data analysis.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	3	1	-	-	-	-	-	-	-	2	-
CO2	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO3	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO4	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	3	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. Principles of Distributed Database Systems, 4th Edition, Özsu, M. Tamer, Valduriez, Patrick, Springer
2. Designing Distributed Systems, Brendan Burns, O'Reilly

❖ Reference book:

1. Distributed Systems An Algorithmic Approach, Second Edition, Sukumar Ghosh, CRC Press
2. Designing Data-Intensive Applications, Martin Kleppmann, O'Reilly

❖ Web material:

1. Coursera: Parallel, Concurrent, and Distributed Programming in Java Specialization
2. <https://hadoop.apache.org/>

❖ Software:

1. Hadoop
2. Oracle Distributed Link

CE379: IMAGE PROCESSING AND COMPUTER VISION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Image fundamental	04
2.	Image formation	06
3.	Image processing	12
4.	Feature extraction	08
5.	Image segmentation	05
6.	Pattern Analysis	10
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Image fundamental	04 Hours	09%
	Elements of visual perception, Human Eye Structure, Digital images, Image Acquisition Techniques, Image sampling and quantization, Spatial & Grey level resolution, Image attributes, Pixels Neighbourhood Adjacency and Distance measures		
2.	Image formation	06 Hours	13%
	The digital camera, Data types and 2d representation of digital images, Geometric primitives, 2D and 3D transformation, 3d Rotation, 2D and 3D projection, lens distortion, Lighting, Reflectance and shading, Sampling and aliasing, Color, Compression		

3.	Image processing	12 Hours	27%
	Pixel transformation, Color transformation, Composition and matching, Histogram processing, Linear filtering, Non-linear filtering,		
4.	Feature extraction	08 Hours	18%
	Edges – canny, LOG, DOG, Line detectors (Hough transformation) Corners – Harries and Hessian affine, Orientation Histogram, SIFT, SURF, HOG, GLOH		
5.	Image segmentation	05 Hours	11%
	Region growing, Edge based approaches to segmentation, Graph-cut, Mean-shift, MRFs, texture segmentation		
6.	Pattern Analysis	10 Hours	22%
	Basics of probability and statistics, Clustering – K-means, K-Medoids, Mixture of Gaussians Classification – Discriminant function, Supervised, Un-supervised, Semi-supervised Classifiers – Bayes, KNN, ANN Dimensionality reduction – PCA, LDA, ICA		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Examine different types of images, general terminology and fundamental concepts of digital image processing and image formation.
CO2	Apply different image processing methods.
CO3	Apply different algorithms for feature extraction.
CO4	Interpret, compare and evaluate segmentation techniques.
CO5	Apply different datamining and mathematical techniques for pattern analysis.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	1	1	-	-	-	-	-	2	-	1	-	-
CO2	2	3	1	2	1	-	-	-	1	-	-	-	2	1
CO3	2	3	1	2	1	-	-	-	1	-	-	-	2	1
CO4	2	3	1	2	1	-	-	-	1	-	-	-	2	1
CO5	2	3	1	2	1	-	-	-	1	-	-	-	2	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. “Digital Image Processing”, Rafael C Gonzalez, Richard E Woods, 2nd Edition, Pearson Education 2003.
2. “Computer Vision a Modern approach”, David A. Forsyth, End Edition, Pearson 2012
3. “Computer Vision Algorithms and Applications”, Richard Szeliski, Springer 2011

❖ Software:

4. MATLAB
5. OpenCV

CE383: BLOCKCHAIN TECHNOLOGY (PE- II)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Basic Cryptography

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Fundamentals Behind Blockchain	04
2.	Blockchain Overview	07
3.	Blockchain Consensus Mechanisms	06
4.	Introduction to Cryptocurrency and Bitcoin	04
5.	Solidity Essential Basic and Advanced	11
6.	Decentralized Application(dapp)	09
7.	WEB3 and Blockchain Tokens	04
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Fundamentals Behind Blockchain	04 Hours	9%
	Public key cryptography, Digital Signature, Cryptographically Secured Hash Functions, Cryptographically Secured Chain of Blocks, Centralized, Decentralized, Distributed Peer to Peer Systems.		
2.	Blockchain Overview	07 Hours	16%

	Evaluation of Blockchain Technology, The History of Blockchain and Bitcoin, Types of Blockchain, Block Structure, Block Header, Merkle Trees, Distributed Ledger, Type of forks in Blockchain.		
3.	Blockchain Consensus Mechanisms	06 Hours	13%
	Proof of Work, Proof of Stake, Delegated Proof of Stack, Proof of Authority, Proof of Elapsed time, Practical Byzantine fault tolerance algorithm, Liquid Proof of Stack		
4.	Introduction to Cryptocurrency and Bitcoin	04 Hours	9%
	Peer to Peer Cash transfer, Digital keys and addresses, Transactions, Mining, The Bitcoin network, Crypto wallets, Types of crypto wallets, Bitcoin Script.		
5.	Solidity Essential Basic and Advanced	11 Hours	24%
	Introduction to Solidity, Data Types, Variables and units, Functions (view, constant, pure, address and fallback), Constructor, Mappings, Modifiers, Structs, and Control Structure & Contract, Emit and Events for logging, Generating ERC-20 Tokens, Deploying the ERC-20 Token Contract. Contract deployment and debugging.		
6.	Decentralized Application(dapp)	09 Hours	20%
	Introduction to dapp and its structure. Ehtereum dapp: Ethereum Blockchain, EVM, Test Network, Application Development Lifecycle, Frameworks and tools. Hyperledger dapp: Hyperledger Fabric and its structure, Chaincode and deployment of network.		
7.	WEB3 and Blockchain Tokens	04 Hours	9%
	NFTs, ERC721, ERC1155, DAO, Defi		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the Basic Cryptography behind the Blockchain Technology and Bitcoin.
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CO2	Define the structure of a Blockchain and classify why and when it is better than a simple distributed database.
CO3	Analyze the consensus mechanisms in a Blockchain Technology and critically assess its applicability in Blockchain based application.
CO4	Analyze to what extent smart and self-executing contracts can benefit automation, governance and transparent environment.
CO5	Design decentralized distributed application and measure the performance of Blockchain against centralized system.
CO6	Attain awareness of the new concept in term of programming that exist in monetizing businesses around Blockchain.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	-	-	1	-	-	2	1	-	-	-	2	-
CO2	-	2	-	-	-	2	1	1	1	-	2	-	1	2
CO3	1	3	2	1	-	2	-	-	2	-	1	2	3	-
CO4	1	1	2	2	3	2	2	2	2	-	3	2	3	1
CO5	1	3	3	2	3	2	2	2	3	2	3	2	3	2
CO6	1	3	1	3	-	2	2	-	3	1	3	2	3	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3:

Substantial (High) If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Imran Bashir, Mastering Blockchain, Packt Second Edition, 2018.
2. Andreas M. Antonopoulos, Mastering Bitcoin, O'Reilly, Second Edition

❖ Reference book:

1. Samanyu, Blockchain Developer's Guide, Packt, 2018.
2. Mark Watney, Blockchain for Beginners: The Complete Step by Step Guide to Understanding Blockchain Technology, July, 2017.

3. Don Tapscott, Blockchain Revolution: How the Technology Behind Bitcoin Is Changing Money, Business, and the World, Hardcover, May 2016.
4. Xun (Brian) Wu, Hyperledger Cookbook, Packt.
5. Mayukh Mukhopadhyay, Ethereum Smart Contract Development, Packt, 2018.

CE359: SOFTWARE GROUP PROJECT -IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	4	-	4	2
Marks	-	100	-	100	

Pre-requisite courses:

- Programming Knowledge

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on a weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - Project Synopsis
 - Software Requirement Specification
 - SPMP
 - Final Project Report
 - Project Setup file with Source code
 - Project Presentation (PPT)

- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 60

Total hours : 60

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Prepare team formation strategies and stages leading to the development of high performing, self-managing teams.
CO2	Identify and define the computing requirements of a problem to propose its appropriate solution.
CO3	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO4	Apply engineering and management principles to achieve project goal.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Recognition of the need for and an ability to engage in continuing professional development.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	2	-	-	-	-	-	-	3	-	-	1	-	2
CO2	-	3	1	2	-	-	-	-	-	-	-	-	2	-
CO3	3	-	2	-	1	-	-	-	-	-	-	-	2	-
CO4	3	-	-	2	-	-	1	-	-	-	-	-	3	-
CO5	-	-	-	-	2	-	-	-	-	1	3	-	-	2
CO6	-	-	-	-	-	1	-	2	-	-	-	3	-	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Books, Magazines & Journals of related topics

❖ Web material:

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

❖ Software:

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS

HS132.02 A: CONTRIBUTORY PERSONALITY DEVELOPMENT

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

- Communication and Soft Skills

Objectives of the Course:

- Become familiar with basic concept of personality and personality development
- Understand personality development theories and strategies
- Evaluate one's personality and inculcate traits of an assertive personality
- Develop an assertive personality
- Develop life skills and required management traits
- Enhance contributory personality for academic and career success

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Concept of Personality	06
2.	Soft Skills and Personality Development	08
3.	Developing Contributory Personality	06
4.	Life skills and Personality Development	06
5.	Contemporary Issues in CPD	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	Concept of Personality	06 Hours	20%
	Meaning of Personality, Types of Personality, Factors contributing to Personality, Personality Traits, Personality Profiling		
2.	Soft Skills and Personality Development	08 Hours	26%
	Positive Thinking and Mind Set, Leadership, Assertiveness		

	and Negotiation Skills, Self-Management, Interpersonal Skills, Being a Team Player		
3.	Developing Contributory Personality	06 Hours	20%
	Concept of Contributory Personality, Characteristics of a Contributor, The Contributor's Vision of Success & Career, The Scope of Contribution in a field, Embarking on the Journey to Contributor ship, Developing Contributor Personality, Reviewing Some Contributors Personalities		
4.	Life skills and Personality Development	06 Hours	20%
	Concept of life skills, Self-awareness, Empathy, Decision Making, Problem Solving		
5.	Contemporary Issues in CPD	04 Hours	14%
	Contemporary Trends and Practices in Contributory Personality Development, Case Study & Presentations		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify one's individual personality strengths and challenges.
CO2	Develop more assertive and optimist attitude towards work and life.
CO3	Develop quintessential soft skills to groom one's personality.
CO4	Identify traits of contributor personality.
CO5	Contribute to self, society, nation, and globe.
CO6	Develop skills of global citizenship to perform societal responsibilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	1	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	1	-	-	-	-
CO3	-	-	-	-	-	1	-	-	3	1	2	1	1	-
CO4	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO5	-	-	2	-	-	2	2	-	-	-	-	-	-	-
CO6	-	-	-	-	-	-	2	2	-	-	-	-	-	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Personality Development & Soft Skills, Oxford University Press
2. Soft Skills, Bookboon
3. Personality Development, Swami Vivekananda; Advaita Ashrama

❖ **Reference book:**

1. Contributor Personality Program Workbook (Volume 1,2),
2. Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd

❖ **Web material:**

1. <https://www.coursera.org/learn/wharton-success>
2. <https://www.coursera.org/learn/personality-types-at-work>
3. <https://www.coursera.org/learn/self-awareness>

B. Tech. (Computer Engineering) Programme

SYLLABI (Semester – 7)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE442: DESIGN OF LANGUAGE PROCESSOR

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	-	6	5
Marks	100	50	-	150	

Pre-requisite courses:

- Digital Electronics
- Operating System
- Theory of Computation

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Overview of Language Processors & Translators	10
2.	Introduction To Compilers	20
3.	Symbol-Table Management	05
4.	Static & Dynamic Memory Allocation & Memory management	06
5.	Semantic Analysis & Intermediate Code Generation	05
6.	Code Optimization	07
7.	Code Generation	07
	Total hours (Theory) :	60
	Total hours (Lab) :	30
	Total hours :	90

Detailed Syllabus:

1.	Overview of Language Processors & Translators	10 Hours	20%
	Language processing activities, fundamental of language processing, Operating System, Interpreter vs compiler Pre-processor & Macro Processors – Subroutine, Macro definition and call, Macro expression, nested macro call, Advanced macro facilities, design of macro pre-processor Loaders - Compile & go, Absolute, Bootstrap, Relocating, Linking loader Linkers - Relocation of Linking Concept, Design of Linker,		

	Linker for MS DOS, Linking for overlays, Linkage editor Assemblers - Elements of Assembly Language Programming, Assembly Scheme, Single pass Assembler, Two pass assembler, Data structure of Assemblers		
2.	Introduction To Compilers	20 Hours	35%
	Pass and Phases of Compiler, grouping of phases, Compiler Contraction tools Lexical analyser - Roles of lexical analyser, input buffering, tokens, Regular Expression, finite Automata Syntax analyser - Context free grammar, Ambiguous grammar, Top-down parsing, Bottom-up parsing, LEX, YACC		
3.	Symbol-Table Management	05 Hours	05%
	Data structures to implement symbol table, Symbol Attributes, Symbol-Table entries, Local Symbol Table management, Global Symbol Table Structure, Storage bindings and Symbolic Registers		
4.	Static & Dynamic Memory Allocation & Memory management	06 Hours	09%
	Data descriptors - Static and Dynamic storage allocation – Storage allocation and access in block structured programming languages – Array allocation and access- Compilation of expressions – Handling operator priorities – Intermediate code forms for expressions – code generator., Register Usage, The run-time stack, Parameter passing disciplines, Code sharing and Position-Independent code		
5.	Semantic Analysis & Intermediate Code Generation	05 Hours	09%
	Intermediate Languages, Declarations, Assignment Statements, Boolean Expressions, Case Statements, Back patching, Procedure Calls		
6.	Code Optimization	07 Hours	11%
	The Principal Sources of Optimization, Optimization of Basic Blocks, Loops in Flow Graphs, Introduction to Global Data-Flow Analysis, Iterative Solution of Data-Flow Equations,		

	Code-Improving Transformations, Dealing with Aliases, Data-Flow Analysis of Structured Flow Graphs, Efficient Data-Flow Algorithms, A Tool for Data-Flow Analysis, Estimation of Types, Symbolic Debugging of Optimized Code		
7.	Code Generation	07 Hours	11%
	Issues in the Design of a Code Generator, The Target Machine, Run-Time Storage Management, Basic Blocks and Flow Graphs, Next-Use Information, A Simple Code Generator, Register Allocation and Assignment, The DAG Representation of Basic Blocks, Peephole Optimization, Generating Code from DAGs, Dynamic Programming Code-Generation Algorithm, Code-Generator Generators		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand design and processing of different language processor, loaders and linkers
CO2	Design top-down and bottom-up parsers
CO3	Identify different memory management schemes of language processors
CO4	Develop semantic analysis scheme to generate intermediate code
CO5	Apply different code optimization techniques
CO6	Develop algorithms to generate code for a target machine

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	1	-	-	3	-	-	-	-	-	-	-	-	-	-
CO3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	2	2	-	2	-	-	-	-	-	-	-	-	-
CO5	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	2	-	-	1	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Alfred Aho, Ravi Sethi, Jeffrey D Ullman, “Compilers Principles, Techniques and- Tools”, Pearson Education Asia.
2. M. Dhamdhare, “System Programming and Operating Systems”, Tata McGraw-Hill.
3. Steven S. Muchnick. Advanced Compiler Design and Implementation

❖ Reference book:

1. Allen I. Holub “Compiler Design in C”, Prentice Hall of India.
2. C. N. Fischer and R. J. LeBlanc, “Crafting a compiler with C”, Benjamin Cummings.
3. J.P. Bennet, “Introduction to Compiler Techniques”, Second Edition, Tata McGraw-Hill
4. HenkAlblas and Albert Nymeyer, “Practice and Principles of Compiler Building with C”, PHI.
5. Kenneth C. Loudon, “Compiler Construction: Principles and Practice”, Thompson Learning.
6. Compiler Construction by Kenneth. C. Loudon, Vikas Pub

❖ Web material:

1. <http://compilers.iecc.com/crenshaw>
2. <http://www.compilerconnection.com>
3. <http://dinosaur.compilertools.net>
4. <http://pltplp.net/lex-yacc>

❖ Software:

1. LEX
2. YACC

CE443: CLOUD COMPUTING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Operating System
- Networking

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Cluster Computing, Grid Computing Systems and Resource Management at Glance	08
2.	Fundamental of Virtualization	06
3.	Fundamental Concepts and Models	06
4.	Cloud-Enabling Technology	05
5.	Fundamental Cloud Architectures	07
6.	Advanced Cloud Architectures	08
7.	Implementation of Cloud	05
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Cluster Computing, Grid Computing Systems and Resource Management at Glance	08 Hours	18%
	Introduction, Eras of Computing, Scalable Parallel Computer Architectures, Towards Low Cost Parallel Computing and Motivations , A Cluster Computer and its Architecture, Clusters Classification, Commodity Components for Clusters, Grid Architecture and Service Modelling, Grid Projects and Grid Systems Built, Grid Resource Management and Brokering, Software and Middleware for Grid Computing,		

	Grid Application Trends		
2.	Fundamental of Virtualization	06 Hours	12%
	Type of Virtualization, Virtualization Technologies, Virtualizes your Environment, Managing Virtualization Environment, Storage Virtualization, Dockers		
3.	Fundamental Concepts and Models	06 Hours	15%
	Roles and Boundaries, Cloud Characteristics, Cloud Delivery Models, Cloud Deployment Models		
4.	Cloud-Enabling Technology	05 Hours	14%
	Broadband Networks and Internet Architecture, Data centre Technology, Virtualization Technology, Web Technology, Multitenant Technology, Service Technology		
5.	Fundamental Cloud Architectures	07 Hours	16%
	Workload Distribution Architecture, Resource Pooling Architecture ,Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture		
6.	Advanced Cloud Architectures	08 Hours	17%
	Hypervisor Clustering Architecture ,Load Balanced Virtual Server Instances Architecture, Non-Disruptive Service Relocation Architecture, Zero Downtime Architecture ,Cloud Balancing Architecture ,Resource Reservation Architecture, Dynamic Failure Detection and Recovery Architecture		
7.	Implementation of Cloud	05 Hours	08%
	Study of Cloud computing Systems like Amazon EC2 and S3, Google App Engine, and Microsoft Azure, Build Private/Hybrid Cloud using open source tools, Deployment of Web Services from Inside and Outside a Cloud Architecture. MapReduce and its extensions to Cloud Computing, HDFS, and GFS		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Assess and examine advantages and disadvantages of virtualization technology.
CO2	Compose services in a distributed computing environment to achieve tasks relevant to a knowledge-based business or public service
CO3	Evaluate a set of business requirements to determine suitability for a cloud computing delivery model.
CO4	Explore the various cloud computing architectures and paradigms.
CO5	Deployment of cloud and identify security implications in cloud computing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	1	0	1	-	-	-	1	2	-
CO2	3	3	3	1	3	2	1	-	1	-	1	3	2	1
CO3	3	3	2	3	3	1	2	-	2	1	-	3	2	1
CO4	3	2	1	2	2	-	-	2	1	-	-	2	2	2
CO5	3	2	3	1	3	2	1	1	2	1	-	3	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text Books:

1. Thomas Erl, Zaigham Mahmood, and Ricardo Puttini, “Cloud Computing Concepts, Technology & Architecture”, Prentice Hall
2. Kai Hwang, Geoffrey C., “Distributed and Cloud Computing”, Morgan Kaufmann is an imprint of Elsevier
3. Navin Sabharwal, Ravi Shankar “Apache CloudStack Cloud Computing” PACKT Publishing

❖ Reference Books:

1. Ravi Shankar, Navin Sabharwa “Cloud Computing First Steps: Cloud Computing for Beginners” Create Space Independent Publishing Platform
2. Rajkumar Buyya, James Broberg, Andrzej Goscinski “Cloud Computing: Principles and Paradigms” Wiley
3. Judith Hurwitz, Robin Bloor “Cloud Computing For Dummies” , for Dummies

❖ **Web material:**

1. <http://www.console.cloud.google.com>
2. <http://www.qwicklabscom>
3. <http://codelabs.developers.google.com>
4. <http://www.docker.com>

❖ **Software/Platform:**

1. NetBeans
2. Eclipse
3. .NET
4. GCP
5. Amazone
6. Microsoft

CE444: INTERNET OF THINGS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Networking
- Embedded systems

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction and Overview of IoT	6
2.	Protocols and Access Technologies (Data link Layer) in IoT	9
3.	Network Layer in IoT	9
4.	Application Layer protocols in IoT	8
5.	Security in IoT	8
6.	IoT Case Studies	5
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Introduction and Overview of IoT	6 Hours	14%
	Origin, Definition, Characteristics, applications, need and scope of IoT, functional stack, Cisco IoT Architecture, Processors and Operating Systems for resource constrained devices, Sensors and actuators, smart objects, IoT vs M2M, IoT vs WoT, IoE.		
2.	Protocols and Access Technologies (Data link Layer) in IoT	9 Hours	20%
	IEEE802.15.4, ZigBee, NFC, Bluetooth,Z-Wave,LPWAN, Evolution of IoT in 1G/2G/3G/4G/5G , IoT Gateways (OAS Platform,Azure IoT Gateway, AWS IoT Gateway,Cisco IoT Gateway,LoRaWAN Gateway, Intel IoT Gateway.)		
3.	Network Layer in IoT	9 Hours	20%

	IPv4, IPv6: IPv6 structure, addressing, routing, interconnecting issues, 6LoWPAN: forwarding, addressing, header compression, neighbor discovery. SDN, Integrating IOT in SDN, IoT/M2M Network Architecture: Sensor Network Protocol, Low energy adaptive clustering hierarchy (LEACH), Sensor protocols for information via negotiation (SPIN). Ad-hoc on-demand distance vector routing system (AODV)		
4.	Application Layer protocols in IoT	8 Hours	17%
	HTTP, CoAP, MQTT, AMQP, XMPP, Integrating Internet Services with Interoperable data encoding with XML, REST Full API, JSON, The Sensor Mark-up Language (SENML)		
5.	Security in IoT	8 Hours	17%
	Challenges in IoT security (Hardware & Software), Attack specific to IoT layers, Hardware Security Attacks, Provisions for securing IoT network, Cisco IoT/M2M Security Framework, TLS, 802.1AR, 802.1X Security Protocol		
6.	IoT Case Studies	5 Hours	12%
	Smart Vehicles, Smart Cities, Smart Homes, Healthcare, Agriculture (Precision farming), Energy Conservation, Supply Chain and Logistics, Industry 4.0 or Industrial IoT IoT Cloud platforms : AWS IoT Platform, Azure IoT Platform, IBM Bluemix Platform, Sensor-Cloud		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Interpret the design issues of IoT & M2M
CO2	Understand the protocol stack specific to sensor network and IoT applications
CO3	Demonstrate a critical understanding of key technologies in data transmission and data processing for IoT.
CO4	Evaluation of privacy management, challenges in designing and securing an IoT system.
CO5	Illustrate the application of IoT in industrial automation and identify real world design Constraints.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	3	2	1	-	-	-	-	-	-	-	3	-
CO3	1	2	3	-	1	-	-	-	-	-	-	-	2	-
CO4	-	1	2	1	1	-	-	-	-	-	-	-	2	-
CO5	2	1	3	-	1	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. Internet of Things: Architectures, Protocols and Standards 1st Edition , Simone Cirani, Gianluigi Ferrari, Marco Picone, and Luca Veltri.
2. Internet of Things: principles and paradigms, Buyya, Rajkumar and Amir Vahid Dasterdji (eds.), Morgan Kaufmann, 2016.
3. From Machine-to-Machine to the Internet of Things: introduction to a new age of intelligence, Holler, Jan et al, Academic Press, 2014.

❖ Reference book:

1. The Internet of Things Enabling Technologies, Platforms, and Use Cases, Pethuru Raj Anupama C. Raman, 2017
2. Building Internet of Things with the Arduino, Doukas, Charalampos, Create Space Independent Publishing Platform, 2012.
3. Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013.

❖ Web material:

1. <http://web.mit.edu/professional/digital-programs/courses/IoT/phone/index.html>
2. https://swayam.gov.in/nd1_noc19_cs65/preview

3. <https://www.edureka.co/blog/iot-tutorial/>
4. <http://www.steves-internet-guide.com/internet-of-things/>

❖ **Software:**

1. Contiki OS
2. Node-Red
3. Proteus
4. Thinker Cad

CE449: BIG DATA ANALYTICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	4	4
Marks	100	50	-	150	

Pre-requisite courses:

- Linux Operating System
- Database Management System

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Big Data and Analytics	02
2.	Introduction to Hadoop and Hadoop Architecture	07
3.	HDFS, HIVE AND HBASE	08
4.	Apache Spark	10
5.	Spark SQL and Spark Streaming	10
6.	Graph Analytics and Data Visualization	08
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Big Data and Analytics	02 Hours	10%
	Introduction to Big Data, Big Data Characteristics, Types of Big Data, Traditional Versus Big Data Approach, Technologies Available for Big Data, Infrastructure for Big Data, Use of Data Analytics, Big Data Challenges.		
2.	Introduction to Hadoop and Hadoop Architecture	07 Hours	15%
	Big Data – Apache Hadoop & Hadoop EcoSystem, Moving Data in and out of Hadoop – Understanding inputs and outputs in Hadoop, Data Serialization		

3.	HDFS, HIVE AND HBASE	08 Hours	20%
	HDFS-Overview, Installation and Shell, Hive Architecture and Installation, Comparison with Traditional Database, HiveQL Querying Data, Sorting And Aggregating, Map Reduce Scripts, Joins & Sub queries, HBase concepts, Advanced Usage, Schema Design, Advance Indexing		
4.	Apache Spark	10 Hours	20%
	Introduction to Data Analysis with Spark, Downloading Spark and Getting Started, Apache Spark components and API stack, Application and Spark Session, Introduction to RDD, RDD and Data Frames.		
5.	Spark SQL and Spark Streaming	10 Hours	20%
	Big Data and Spark SQL, Spark-Managed Tables, Reading Tables into Data Frames, Aggregations, Joins, Creating Views, Spark Streaming and Challenges of Stream Processing, Spark's Streaming APIs, Spark streaming case study.		
6.	Graph Analytics and Data Visualization	08 Hours	15%
	Apache Spark GraphX: Property Graph, Graph Operator, SubGraph, Triplet, Neo4j: Modeling data with Neo4j, Cypher Query Language: General clauses, Read and Write clauses. Big Data Visualization with Power BI, Apache Super-Set		

At the end of the course, the students will be able to

CO1	Understand the key issues in big data management and its associated applications in intelligent business and scientific computing
CO2	Acquire fundamental enabling techniques and scalable algorithms like Hadoop, Map Reduce, Hive and Spark in big data analytics.
CO3	Evaluate and apply appropriate principles, techniques and theories to large-scale data science problems using various databases with analytics and visualizations.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1	-	-	-	-	-	-	-	-	-	1	1
CO2	1	2	3	1	3	-	-	-	-	-	-	-	2	-
CO3	-	1	3	3	3	-	-	-	-	-	-	-	2	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Bart Baesens , Analytics in a Big Data World: The Essential Guide to Data Science and its Applications, ,Wiley, 2014
2. Jules S. Damji, Learning SparkLightning-Fast Data Analytics, O’Reilly Media Inc, 2020.
3. Spark: The Definitive Guide by Bill Chambers and Matei Zaharia, O’Reilly Media Inc. 2018.

❖ Reference book:

1. Xyz Dirk Deroos et al., Hadoop for Dummies, Dreamtech Press, 2014.
2. Chuck Lam, Hadoop in Action, December, 2010.
3. Leskovec, Rajaraman, Ullman, Mining of Massive Datasets, Cambridge University Press.
4. I.H. Witten and E. Frank, Data Mining: Practical Machine learning tools and techniques.

❖ Web material:

1. <https://cognitiveclass.ai/>
2. <https://codelabs.developers.google.com/>

❖ Software & Platform:

1. R & SPSS
2. Hadoop, HBase, Hive, Pig, Spark
3. Casandra, Neo4j, NoSQL

CE474: BLOCKCHAIN TECHNOLOGY (PE- III)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Basic Information Security

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
8.	Fundamentals Behind Blockchain	07
9.	Blockchain Overview	07
10.	Block-Chain Consensus Mechanisms	06
11.	Introduction to BitCoin	06
12.	Solidity Essential	08
13.	Decentralized Application	08
14.	Blockchain Research issues, challenges and use cases	03
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Fundamentals Behind Blockchain	07 Hours	11%
	Symmetric key cryptography, Public key cryptography, Digital Signature, Cryptographically Secured Hash Functions, Cryptographically Secured Chain of Blocks, Merkle Trees.		
2.	Blockchain Overview	06 Hours	10%
	Evaluation of Blockchain Technology, Distributed Systems, The History of Blockchain and Bitcoin, Types of Block-		

	Chain		
3.	Block-Chain Consensus Mechanisms	12 Hours	20%
	Practical Byzantine fault tolerance algorithm, Proof of Work, Proof of Stake, Proof of Authority, Proof of Elapsed time		
4.	Introduction to BitCoin	12 Hours	20%
	Digital keys and addresses, Transactions, Mining, The Bitcoin network, Wallets, Bitcoin payments		
5.	Solidity Essential	08 Hours	13%
	Using the Remix IDE, Data Types and Functions, Creating Inline Assembly Functions, Mappings, Modifiers, Structs, and More, Generating ERC-20 Tokens, Extending Token Security (ERC-223), Deploying the ERC-20 Token Contract.		
6.	Decentralized Application	08 Hours	17%
	DAPP using Ehtereum, DAPP using Hyperledger		
7.	Blockchain Research issues, challenges and use cases	05 Hours	09%
	KYC case study, Land Registry, Supply chain.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the Basic Cryptography behind the Blockchain Technology and Bitcoin.
CO2	Define the structure of a Blockchain and classify why and when it is better than a simple distributed database.
CO3	Analyse the consensus mechanisms in a Blockchain Technology and critically assess its applicability in Blockchain based application.
CO4	Analyse to what extent smart and self-executing contracts can benefit automation, governance and transparent environment.
CO5	Design decentralized distributed application and measure the performance of Blockchain against centralized system.
CO6	Attain awareness of the new challenges that exist in monetizing businesses around Blockchain.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	-	-	1	-	-	2	1	-	-	-	2	-
CO2	-	2	-	-	-	2	1	1	1	-	2	-	1	2
CO3	1	3	2	1	-	2	-	-	2	-	1	2	3	-
CO4	1	1	2	2	3	2	2	2	2	-	3	2	3	1
CO5	1	3	3	2	3	2	2	2	3	2	3	2	3	2
CO6	1	3	1	3	-	2	2	-	3	1	3	2	3	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. Imran Bashir, Mastering Blockchain, Packt Second Edition, 2018.
2. Andreas M. Antonopoulos, Mastering Bitcoin, O'Reilly, Second Edition

❖ Reference book:

1. Samanyu, Blockchain Developer's Guide, Packt, 2018.
2. Mark Watney, Blockchain for Beginners: The Complete Step by Step Guide to Understanding Blockchain Technology, July, 2017.
3. Don Tapscott, Blockchain Revolution: How the Technology Behind Bitcoin Is Changing Money, Business, and the World, Hardcover, May 2016.
4. Xun (Brian) Wu, Hyperledger Cookbook, Packt.
5. Mayukh Mukhopadhyay, Ethereum Smart Contract Development, Packt, 2018.

❖ Web material:

1. <https://medium.com/topic/blockchain>

❖ Software:

1. Ethereum
2. Hyperledger Fabric 1.4
3. Truffle
4. Solidity

CE475: WIRELESS COMMUNICATION & MOBILE COMPUTING
(PE - III)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Basic Knowledge of Computer Networks

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Wireless Communication Fundamentals	02
2.	Telecommunication Systems	10
3.	Wireless LAN	13
4.	Mobile Network Layer	10
5.	Transport and Application Layer	08
6.	Data Dissemination and Management	02
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Wireless Communication Fundamentals	02 Hours	5%
	Introduction, Wireless transmission – Frequencies for radio transmission – Signals – Signal Propagation, Multiplexing – Modulations – Spread spectrum, MAC – SDMA – FDMA – TDMA – CDMA – Cellular Wireless Networks		
2.	Telecommunication Systems	10 Hours	25%
	GSM: Mobile services, System architecture, Radio interface, Protocols, Localization And Calling, Handover, Security, New data services; DECT: System architecture, Protocol architecture; TETRA, UMTS and IMT-2000		
3.	Wireless LAN	13 Hours	30%
	Wireless LAN – IEEE 802.11 - Architecture – services – MAC – Physical layer, IEEE 802.11a - 802.11b – 802.11n		

	standards, Bluetooth, Hyperlan, Wi-Fi, WiMax - Overview		
4.	Mobile Network Layer	10 Hours	20%
	Mobile IP, Dynamic Host Configuration Protocol, Routing Protocols – DSDV – DSR – Alternative Metrics		
5.	Transport and Application Layer	08 Hours	15%
	Traditional TCP, Classical TCP improvements – WAP, WAP 2.0		
6.	Data Dissemination and Management	02 Hours	05%
	Challenges, Data dissemination, Mobile data replication, Mobile data caching, Mobile cache maintenance, mobile web caching, caching in ad hoc networks.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Classify the fundamental concepts of Wireless Networks and its access techniques.
CO2	Understand and identify the GSM, GPRS and Bluetooth software model for mobile computing.
CO3	The ability to develop applications that are mobile-device specific and demonstrate current practice in mobile computing contexts.
CO4	Understanding of the characteristics and limitations of mobile hardware devices including their user-interface modalities.
CO5	Able to promote the awareness of the life-long learning, business ethics, professional ethics and current marketing scenarios in wireless network.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	-	-	-	-	-	-	-	-	-	1	1
CO2	1	1	2	2	1	-	-	-	-	-	-	-	-	1
CO3	2	3	2	1	2	-	-	-	-	-	-	-	-	2
CO4	2	2	1	1	1	-	-	-	-	-	-	-	2	2
CO5	1	2	2	1	1	-	-	-	-	-	-	2	1	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text Books:

1. “Mobile Computing: Technology, Applications and Service Creation” by Asoke K Talukder and Roopa R Yavagal, TMH, ISBN: 0-07-058807-4

❖ Reference Materials:

1. Jochen Schiller, “Mobile Communications”, PHI/Pearson Education, Second Edition, 2003.
2. William Stallings, “Wireless Communications and Networks”, PHI/Pearson Education, 2002.
3. Kaveh Pahlavan, Prasanth Krishnamoorthy, “Principles of Wireless Networks”, PHI/Pearson Education, 2003.
4. Uwe Hansmann, Lothar Merk, Martin S. Nicklons and Thomas Stober, “Principles of Mobile Computing”, Springer, New York, 2003.
5. Hazysztof Wesolowski, “Mobile Communication Systems”, John Wiley and Sons Ltd, 2002
6. Research papers from IEEE, Springer etc.

❖ Web Materials:

1. www.ietf.org – For drafts
2. www.ieee.org – For standards and technical research papers

❖ Software:

1. OMNET++
2. SUMO
3. MATLAB
4. NS3

CE476: MACHINE LEARNING (PE - III)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Probability, linear algebra, calculus and programming language

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Fundamental concepts and Statistical Learning Techniques	06
2.	Neural Networks	10
3.	Bayesian Learning	03
4.	Supervised and Unsupervised Learning	08
5.	Reinforcement learning	06
6.	Kernel Methods	06
7.	Deep Neural Networks	06
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Fundamental concepts and Statistical Learning Techniques	06 hours	13%
	Need for Machine Learning. Types of Machine Learning: Supervised Learning, Unsupervised Learning, Reinforcement Learning Statistical Learning Techniques: Linear Regression, Logistic Regression, Multi Linear regression, Correlation, Gradient Descent for Multiple Variables		
2.	Neural Networks	10 Hours	22%
	Neurons and biological motivation. Linear threshold units. Perceptrons: representational limitation and gradient descent		

	training, Perceptron learning rule, Hebbian learning rule, Delta Learning rule, Multilayer networks and Backpropagation Learning Algorithm, Feed Forward, Activation Functioning, Types of Neural Network Architecture.		
3.	Bayesian Learning	03 hours	7%
	Bayes Theorem, Bayes Theorem and Concept Learning, Maximum Likelihood and Least Squared Error Hypothesis, Maximum likelihood hypothesis for Predicting probabilities, Minimum Description Length Principle, Bayes Optimal Classifier, Gibbs Algorithm, Naïve Bayes Classifier.		
4.	Supervised and Unsupervised Learning	08 hours	18%
	Supervised Learning: Classification, Decision Tree, Naïve Bias, Support Vector Machine, Neural Network, K- Nearest Neighbour. Unsupervised Learning: Clustering, Nonparametric Methods, K-means, Hierarchical clustering, Density based clustering		
5.	Reinforcement learning	06 hours	13%
	Q Learning, Non deterministic rewards and Actions.		
6.	Kernel Methods	06 hours	13%
	Support Vector Machine, Sparse kernel machines, Bias-Variance trade-off. Regularization and model/feature selection, Sampling Methods.		
7.	Deep Neural Networks	06 hours	14%
	Introduction to Deep Learning, Deep Neural Network, Convolution Neural Network, Recurrent Neural Network, Transfer learning, Auto Encoders.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Apply basic concepts of Machine Learning and Understanding of standard learning algorithms.
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CO2	Analyse mathematical modelling of various Machine Learning algorithms.
CO3	Understanding challenges of machine learning like data characteristics, model selection, and model complexity.
CO4	Identify strengths and weaknesses of machine learning techniques suitable for a given problem domain and data set.
CO5	Design and implement of various machine learning algorithms in a range of real-world applications.
CO6	Evaluate and interpret the results of learning algorithms.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO2		2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	3	-	-	-	-	-	3	-	-	-	-	-	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-	2	-
CO5	3	-	2	2	2	-	-	-	-	-	2	-	3	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	3	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Xyz Machine Learning, Tom Mitchell, McGraw Hill, 1997. ISBN 0070428077
2. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, 2004

❖ Reference book:

4. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.

5. Richard O. Duda, Peter E. Hart & David G. Stork, "Pattern Classification. Second Edition", Wiley & Sons, 2001.
6. Trevor Hastie, Robert Tibshirani and Jerome Friedman, "The elements of statistical learning", Springer, 2001.
7. Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", MIT Press, 1998.

❖ **Web material:**

1. <https://www.youtube.com/watch?v=f-oHSmB48rY&list=PLKvX2d3IUq586Ic9gIhZj6ubpWV-OJfl4>
2. https://www.youtube.com/watch?v=CS4cs9xVecg&list=PLkDaE6sCZn6Ec-XTbcX1uRg2_u4xOEky0
3. <https://www.youtube.com/watch?v=UzxYlbK2c7E>
4. <https://www.youtube.com/playlist?list=PLAwXTw4SYaPkQXg8TkVdIvYv4HfLG7SiH>

❖ **Software:**

1. Anaconda

CE450: SOFTWARE GROUP PROJECT-V

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	4	-	4	2
Marks	-	100	-	100	

Pre-requisite courses:

- Programming Knowledge

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - g. Project Synopsis
 - h. Software Requirement Specification
 - i. SPMP
 - j. Final Project Report
 - k. Project Setup file with Source code
 - l. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 30

Total hours : 30

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Identify and define the computing requirements of a problem to propose its appropriate solution.
CO2	Correlate knowledge of different subjects and apply it to implement solution of the problem.
CO3	Apply engineering and management principles to achieve project goal.
CO4	Prepare technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO5	Able to communicate effectively with a range of audiences.
CO6	Recognition of the need for and an ability to engage in continuing professional development.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	2	-	-	-	-	2	-	-	-	3	-
CO2	3	-	1	1	3	1	1	-	-	-	-	2	1	1
CO3	-	-	3	-	3	1	1	-	2	-	3	2	3	-
CO4	-	-	-	-	3	-	2	-	2	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	3	-	-	-	-	2	-	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Books, Magazines & Journals of related topics

❖ Web material:

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

❖ Software:

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS

CE446: SUMMER INTERNSHIP-II

Credits and Hours:

Teaching Scheme	Project	Practical	Tutorial	Total	Credit
Hours	90	-	-	90	3
Marks	150	-	-	150	

Objectives of the Course:

- To get familiar with modern tools and technologies use in company/industry/organization
- To get involved in design, development and testing practices followed in the company/industry/organization
- To enhance their soft-skills, presentation skills, interpersonal skills, documentation skills and office etiquettes required to sustain in company/industry/organization environment
- To participate in teamwork and preferably as part of a multi-disciplinary team
- To make them aware about company/industry/organization best practices, processes and regulations.
- To make them more productive, consistent and punctual.

Outline of the Course:

1.	Instructional Method and Pedagogy
	• Summer internship shall be at least 90 hours during the summer vacation only. • Department/Institute will help students to find an appropriate company/industry/organization for the summer internship. • The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline. • Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed. • During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and those of the University. • Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.

	• The student shall submit two documents to the Coordinator for the evaluation of the summer internship: • Summer Internship Report • Summer Internship Assessment Form • Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted through the presentation to the Coordinator by the first day of the new term. • The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester. • If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.
2.	Format of Summer Internship Report
	The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections: • Cover Page • Page of Approval and Grading • Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at. • TABLE OF CONTENTS (with the corresponding page numbers) • LIST OF FIGURES AND TABLES (with the corresponding page numbers)

- **DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION:**
Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- **INTRODUCTION:** In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- **PROBLEM STATEMENT:** What is the problem you are solving, and what are the reasons and causes of this problem.
- **SOLUTION:** In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
- **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it

	and waste tons of paper.	
	Total hours (Project):	90
	Total hours:	90

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Ability to integrate existing and new technical knowledge for industrial application.
CO2	Executing work with team and teammates from other disciplines
CO3	Get practices and experience related to professional and ethical issues in the work environment
CO4	Experience of demonstrating the impact of the internship on their learning and professional development.
CO5	Understanding of lifelong learning processes through critical reflection of internship experiences.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	2	1	-	-	2
CO3	-	-	-	-	-	-	1	3	-	1	-	-	-	1
CO4	-	-	-	-	-		3	1	-	-	-	-	1	-
CO5	-		-	-	-	1	-	-	-	-	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Books, Magazines & Journals of related topics

❖ **Web material:**

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

❖ **Software:**

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS
5. FLUTTER
6. NODE/REACT NATIVE

B. Tech. (Computer Engineering) Programme

SYLLABI (Semester – 8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE451: SOFTWARE PROJECT MAJOR

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	36	36	18
Marks	-	600(250+350)	600	

Pre-requisite courses:

- Software Engineering and other relevant courses, Development tools and languages, soft skill.

Outline of the Course:

- Students work with industry/organization 4-to-6 months for development or research project.
- The definitions are verified and guidance are given to take the project at next level and can help industry, society or environment.
- The external guide (at industry/Organization) and internal guide (Institute faculty) continuously monitor the students' work and project file is maintained per group to document all the measurable work.
- Project work should include whole SDLC/Agile of development of software / hardware system as solution of particular problem by applying principles of Software Engineering.
- Project is evaluated twice during the semester by internal faculty of the university as a part of continuous evaluation.
- Final evaluation at the end of the semester is done by Industry experts or faculties from reputed university.
- Students have to submit SRS, SPMP, Design documents, Code and Test Cases in form of Project report.
- The feedback of the students by external guide and Evaluator are taken during the semester to improve the teaching learning and evaluation process.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify and justify/analyse the requirements of the projects with enhancement of the required tools and technology by individual and team.
CO2	Solve challenging projects for commercial, societal and environment benefit.
CO3	Apply the engineering knowledge to full fill the requirements of the projects pertaining to any discipline.
CO4	Explain the importance of planning, documentation, punctuality and work ethics.
CO5	Document the work which is carried out in proper format with industry standards.
CO6	Showcase the soft skill.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	2	3	-	1	-	3	1	1	1	3	1
CO2	2	-	3	-	-	3	2	-	-	-	-	-	2	2
CO3	3	2	1	1	1	-	-	-	1	-	2	1	3	1
CO4	3	1	2	-	-	-	-	3	-	-	1	2	2	3
CO5	3	-	2	2	-	-	-	-	-	2	-	3	1	1
CO6	1	-	1	1	-	-	-	3	1	3	1	1	1	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material (Suggested by Internal or External guide):

- Reading Materials, web materials, Project reports with full citations.
- Books, magazines & Journals of related topics.
- Various software tools and programming languages compiler related to topic.

❖ Web Materials:

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com